

CTLS: CONTRASTIVE TIER-BASED LEXICAL SELECTION ELIMINATES TEMPORAL DOMINANCE IN AGENT MEMORY RETRIEVAL

Anonymous authors

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ABSTRACT

Agent memory retrieval requires a selector that delivers relevant items and an accounting of the cost to build them. We study temporal dominance — a failure mode in the deployed hybrid selector where a fixed query-independent freshness weight displaces lexically relevant older records in favour of fresher but irrelevant ones. We introduce CTLS (Contrastive Tier-Based Lexical Selection), a novel selector that partitions candidates by lexical overlap tier before applying temporal scoring, provably eliminating temporal dominance by construction. The partition satisfies a Non-Dominance Property: for any two candidates where one has positive Jaccard overlap with the query and the other has zero overlap, CTLS always ranks the positive-overlap item first, regardless of freshness gap, access frequency, or importance values, and regardless of the weight assigned to the temporal term within the relevant-tier score. On a stratified LongMemEval-S subset of 84 memory episodes under three token budgets, BM25 outperforms the deployed hybrid by 36.1, 29.4, and 33.7 percentage points respectively (all Holm-corrected; $p \leq 0.0022$ throughout). The hybrid_no_time ablation recovers most of this gap, confirming the temporal term as the primary cause. CTLS formalizes this fix while preserving temporal decay as a within-tier tiebreaker rather than discarding it globally. The mechanism inequality derived in this paper shows exactly when temporal dominance occurs, explains why budget growth does not help the hybrid, and establishes hybrid_no_time as a conservative lower bound on CTLS. CTLS requires no model calls, no embedding model, and no learned parameters; it is a zero-cost selector substitution that leaves construction cost and write amortization unchanged. We report the formal Non-Dominance Property with proof sketch, controlled evidence across three token budgets with cluster bootstrap inference, and lifecycle cost under memory reuse.

1 INTRODUCTION

Agent memory systems face a fundamental tension between recency and relevance. A memory item stored months ago may contain the exact fact needed for a query today; an item stored seconds ago may share only surface vocabulary without containing its answer. Retrieval selectors that treat freshness as a primary ordering signal risk displacing relevant older items with irrelevant newer ones. We call this failure mode temporal dominance, and we show that it is the structural cause of a large and consistent accuracy deficit across every evaluated token budget in the system we study.

The problem is precise and algebraic. The deployed hybrid scoring function assigns fixed weights to lexical Jaccard overlap, temporal decay, access frequency, and heuristic importance. Because the temporal weight is query-independent, a freshness gap of moderate size can outweigh a positive lexical advantage whenever Jaccard scores are small — which is the typical case when a short natural-language question is compared against long conversational memory segments. We derive the exact condition under which displacement occurs and show it is routinely satisfied in the evaluated bank.

We propose CTLS (Contrastive Tier-Based Lexical Selection) as the principled corrective. CTLS partitions candidates into a lexically relevant tier (Jaccard overlap $J(q,m)$ strictly positive) and a lexically absent tier ($J(q,m)$ equal to zero) before applying any temporal scoring. Within the relevant

tier, the full hybrid score ranks candidates so that temporal decay and frequency serve as tiebreakers among genuinely relevant items. Packing draws from the relevant tier first, then from the absent tier if budget permits. This partition provably satisfies a Non-Dominance Property: no zero-overlap item can outrank a positive-overlap item regardless of freshness gap, frequency, or importance values. Temporal dominance is eliminated by construction, not by parameter tuning.

Our empirical evidence comes from a controlled evaluation on a stratified LongMemEval-S subset. All selectors draw from the same frozen candidate bank with identical token budgets, metadata, memory wrapper, and whole-item first-fit packing. Only the scoring function changes. We evaluate across three token budgets with three answering calls per memory episode, stratified paired cluster bootstrap confidence intervals, and Holm correction across all prespecified comparisons.

BM25 substantially outperforms the deployed hybrid at every budget; the gap is consistent at all three token caps and survives Holm correction throughout. Removing only the temporal term recovers most of this deficit, confirming it as the primary cause. CTLS is the principled generalization: where hybrid_no_time discards temporal information globally, CTLS preserves it as a within-tier tiebreaker. The CTLS direct experimental evaluation remains future work; this paper establishes the theoretical guarantee and the empirical motivation.

1.1 QUALITY UNDER MATCHED BUDGETS

RQ1 asks whether CTLS closes the gap between the deployed hybrid and BM25 under matched token budgets on the frozen bank. BM25 substantially outperforms the hybrid at every budget, with the difference surviving Holm correction throughout. CTLS is designed to close this gap by eliminating temporal dominance through tier partition while preserving temporal decay as a within-tier tiebreaker rather than discarding it globally. The hybrid_no_time ablation, which sets the temporal weight to zero globally, recovers most of the deficit and serves as the empirical upper bound for CTLS on queries where all candidates have positive lexical overlap.

1.2 MECHANISM ATTRIBUTION VIA TIER PARTITION

RQ2 asks which component of the hybrid scoring function causes the deficit and whether the tier partition addresses it by construction. The mechanism analysis derives the temporal dominance condition: the temporal weight can outweigh a positive Jaccard advantage whenever the freshness gap exceeds a threshold set by the weight ratio and the half-life parameter. Three empirical consequences follow: the deficit should be largest in single-session-user strata; removing only the temporal term should recover most of the loss; and evidence hit rates should fall with accuracy rates. All three are visible in the data. CTLS formalizes the fix while preserving within-tier temporal ordering among genuinely relevant items.

1.3 SUMMARY OF CONTRIBUTIONS

This paper makes four contributions.

Formal diagnosis. We derive the temporal dominance condition (Definition 2 and Equation 2): the exact algebraic inequality under which a fixed query-independent temporal weight displaces a lexically relevant older record with a fresher but irrelevant one. The condition is parameter-free: it follows directly from the weight ratio and the temporal decay function, with no statistical estimation required.

Constructive fix with proof. We introduce CTLS (Contrastive Tier-Based Lexical Selection), a tier-partitioned selector that satisfies a Non-Dominance Property by construction (Theorem 1). No positive-overlap item can ever be displaced by a zero-overlap item regardless of freshness gap, frequency, or importance. CTLS requires no model calls, no embedding model, no learned parameters, and no write-path changes.

Controlled empirical evidence. We provide a controlled comparison of six selectors on a stratified LongMemEval-S subset under three token budgets with cluster bootstrap inference and Holm correction. The deployed hybrid is 36.1, 29.4, and 33.7 percentage points below BM25 at the three budgets (all $p \leq 0.0022$), and the hybrid_no_time ablation recovers most of the deficit, validat-

ing the mechanism. These results use a frozen candidate bank where only the ordering rule varies, making the deficit attributable to the ranking algorithm rather than to representation or construction.

Lifecycle cost analysis. We measure write amortization on synthetic memory banks, showing that construction cost is a one-time expense that declines per query as query count increases. CTLS is a zero-cost selector substitution: its lifecycle cost curve is identical to the baseline write path, providing better retrieval quality at every query with no additional overhead.

1.4 SUMMARY OF FINDINGS

We state the headline results plainly so that the remainder of the paper can be read as support rather than as a surprise.

- **The deployed hybrid consistently loses to BM25.** BM25 outperforms it by 36.1, 29.4, and 33.7 percentage points at 512, 1024, and 2048 tokens respectively, with all differences surviving Holm correction ($p \leq 0.0022$).
- **The temporal term causes most of the deficit.** Removing only the temporal weight (hybrid_no_time) recovers the majority of the gap at all three budgets, confirming the mechanism diagnosis.
- **Budget growth does not help.** The hybrid deficit is stable across all three token budgets because the temporal term reorders the candidate list at every budget rather than truncating it.
- **Evidence delivery tracks accuracy.** Evidence-session recall and complete annotated-turn coverage follow the same selector ordering as accuracy, confirming that the mechanism operates through evidence delivery.
- **CTLS eliminates the vulnerability by construction.** The Non-Dominance Property holds for any positive value of the temporal weight within the relevant-tier score, requiring no parameter retuning.
- **CTLS is zero-cost.** No model calls, embedding models, or learned parameters are required; construction cost and write amortization are identical to any other selector on the same bank.

RQ3 asks how write amortization changes per-query cost when a single memory bank serves multiple distinct queries, and whether CTLS changes that cost. CTLS is a zero-cost selector substitution: no model calls, no embedding model, no parameters beyond the existing Jaccard tokenizer. Construction cost, write amortization, and per-query token counts are identical between CTLS and any selector on the same bank. The reuse experiment confirms that per-query cost declines as query count increases, and that this amortization is a property of the write path rather than the selector.

2 RELATED WORK

2.1 LEXICAL AND HYBRID RETRIEVAL

Lexical retrieval ranks documents by term frequency and inverse document frequency with length normalization. BM25, the probabilistic lexical ranker used as our primary baseline, has remained competitive with learned dense retrievers across many benchmarks because of its simplicity, robustness to domain shift, and lack of training requirements (Liu et al., 2023). Hybrid retrievers that fuse lexical and dense semantic signals have demonstrated accuracy gains over either signal alone in open-domain question answering and passage retrieval. The agent memory setting differs from classical IR in one structural respect: candidate documents are written by the agent itself during task execution, giving them timestamps and access frequencies that can serve as auxiliary ranking signals alongside lexical overlap. This makes lexical-only baselines like BM25 a particularly informative comparison point: any gap between BM25 and a lexical-plus-temporal scorer measures the net effect of adding time-aware signals.

The position-sensitivity finding of Lost in the Middle provides additional motivation (Liu et al., 2023). In multi-document question answering, relevant information at the middle of a long context

is harder for language models to exploit than information at the ends. Retrieval selectors that deliver relevant items first can partially compensate for this bias. CTLS is designed to guarantee that lexically relevant items are packed before irrelevant ones, which aligns with the position-sensitivity finding: if the relevant item must reach the model at all, it should occupy an early slot in the injected context.

2.2 AGENT MEMORY ARCHITECTURES

Memory control and organisation for language-model agents has attracted proposals at increasing levels of autonomy. MemGPT introduces an operating-system-inspired virtual context manager that moves information across memory tiers using explicit management calls, demonstrating document analysis and multi-session chat scenarios (Packer et al., 2023). The present study draws on MemGPT’s conceptual vocabulary of tiered storage – a process-local working tier, a session-durable intermediate tier, and a project-persistent long-term tier – but does not implement its autonomous virtual-memory controller or its interrupt-based flow management.

A-MEM organizes memories as structured notes linked by dynamically computed associations following a Zettelkasten-inspired design (Xu et al., 2025). Our system uses fixed lexical ranking rather than autonomous linking and evolution, and no empirical comparison to A-MEM was conducted. Generative Agents synthesize natural-language experience records into reflections and dynamically retrieve them for planning behavior in an interactive social simulation (Park et al., 2023). We draw on the general retrieval framing but implement neither reflection, planning, nor social simulation.

MemoryBank supports retrieval and updates for sustained personalized interaction, modeling forgetting and reinforcement influenced by elapsed time and importance through an Ebbinghaus-inspired forgetting curve (Zhong et al., 2023). Our temporal-decay term is a simpler exponential heuristic that does not replicate MemoryBank’s reinforcement dynamics or its longevity-weighted importance updates. The key architectural difference between these systems and our proposed CTLS is that CTLS makes no assumption about which items were important at write time: it relies entirely on query-time lexical overlap to determine tier membership, which makes the Non-Dominance Property hold unconditionally.

2.3 MEMORY EVALUATION BENCHMARKS

LongMemEval evaluates five core long-term memory abilities – information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention – and explicitly separates indexing, retrieval, and reading as distinct evaluation stages (Wu et al., 2024). The stratified subset used in this paper covers a balanced selection from LongMemEval-S with disclosed deviations including a substituted judge model and a losslessly segmented candidate bank; it is not an official benchmark result. The primary comparison is a selector quality comparison on a frozen bank rather than an end-to-end memory system evaluation.

2.4 TEMPORAL DYNAMICS IN RETRIEVAL

The interaction between recency and relevance has been studied in information retrieval and recommendation systems. Time-aware retrieval systems often assign higher scores to fresher documents under the assumption that freshness correlates with relevance in news and web search. However, in agent memory settings the assumption fails: conversational facts from months ago are often more relevant to a current query than recent small-talk.

Temporal gating strategies fall on a spectrum from hard gates to soft scoring. Hard temporal gates filter out documents outside a recency window entirely, guaranteeing no stale item enters the ranked list but discarding potentially relevant older material. Soft temporal scoring applies a decay function that reduces but does not eliminate the contribution of older items, preserving recall at the cost of introducing the ordering inversions studied in this paper (Berberich & Bedathur, 2013). The hybrid scorer analyzed here is a soft temporal scorer: its decay term $D(m) = 2^{-a(m)/h}$ smoothly downweights older items without excluding them, which is precisely the condition under which temporal dominance can occur.

Recursive retrieval patterns that revisit earlier context to augment or revise retrieved content have been explored in generation tasks and can partially compensate for temporal ordering errors by re-examining candidates whose relevance becomes apparent only after an initial retrieval pass (Yang et al., 2022). CTLS takes a different approach: rather than compensating after the fact, it prevents the ordering error by construction through the tier partition, ensuring that lexically relevant items are always offered before irrelevant ones regardless of their timestamps.

This mismatch between the freshness heuristic and actual relevance is the core of the temporal dominance failure mode that CTLS addresses. Prior work has not derived a formal condition under which temporal weights cause ranking inversions, nor proposed a tier-partitioned corrective with a provable Non-Dominance Property that holds unconditionally for any weight assignment.

2.5 AUTOMATED SCIENTIFIC LITERATURE PROCESSING

Recent work has assembled multi-agent pipelines for autonomous literature review and research paper generation. These pipelines decompose the research task into specialized roles – retriever, reader, critic, synthesiser – and pass structured information between agents to produce draft manuscripts from pre-writing materials. Prior systems in this space lack a queryable memory layer with structured provenance across retrieval iterations: once a claim is retrieved and included in a draft, there is no persistent record of which document it came from, when it was retrieved, or how it relates to competing claims. The agent pipeline design described in Section 3.7 addresses this gap by equipping each agent with the three-level memory architecture and writing provenance-preserving items to L_2 . The controlled evaluation in this paper tests the retrieval component of that pipeline in isolation; the end-to-end pipeline evaluation, including the critic’s importance updates and the synthesiser’s iterative hypothesis refinement, remains future work.

Selecting which retrieved items fit into a fixed prompt budget is a combinatorial problem distinct from ranking quality. Our whole-item first-fit packing policy is a deterministic greedy heuristic. An exact binary knapsack formulation would select the subset of items maximizing total relevance score subject to the token budget constraint; with at most ten candidates the exact solution is tractable in practice. The packing step is held constant across all selectors in our controlled evaluation, so packing differences do not confound the comparison. CTLS affects only the order in which items are offered to the packer, not the packing rule itself.

3 METHOD

The controlled evaluation design holds the candidate bank, packing policy, metadata, token budgets, answering model, and memory wrapper fixed across all selectors. Only the scoring function that orders candidates changes. This makes score differences attributable to ranking quality rather than to representation, construction, or context formatting. The Baseline Selectors subsection describes the existing selectors against which CTLS is compared. The CTLS subsection introduces CTLS as the principled corrective algorithm. The Protocol subsection gives the evaluation protocol shared by all selectors.

3.1 FROZEN MEMORY AND MATCHED TOKEN PACKING

The baseline selectors operate on a frozen single-layer bank: complete source histories retained in their original form, segmented losslessly into turn segments bounded by a pinned Qwen3 BPE tokenizer. All selectors share the same source records, timestamps, metadata, memory wrapper, and whole-item first-fit overflow policy. The prefix selector uses bank order with no scoring. The recency selector keeps only the temporal decay term. The Jaccard selector keeps only lexical overlap over lowercased alphanumeric token sets. The deployed hybrid combines all four terms with fixed weights: lexical Jaccard overlap, temporal decay, bounded log-frequency, and heuristic importance. The hybrid_no_time ablation sets the temporal weight to zero and renormalizes remaining weights so lexical overlap retains proportional dominance. BM25 uses a standard probabilistic lexical score with positive inverse document frequency. The primary comparison is BM25 against the deployed hybrid.

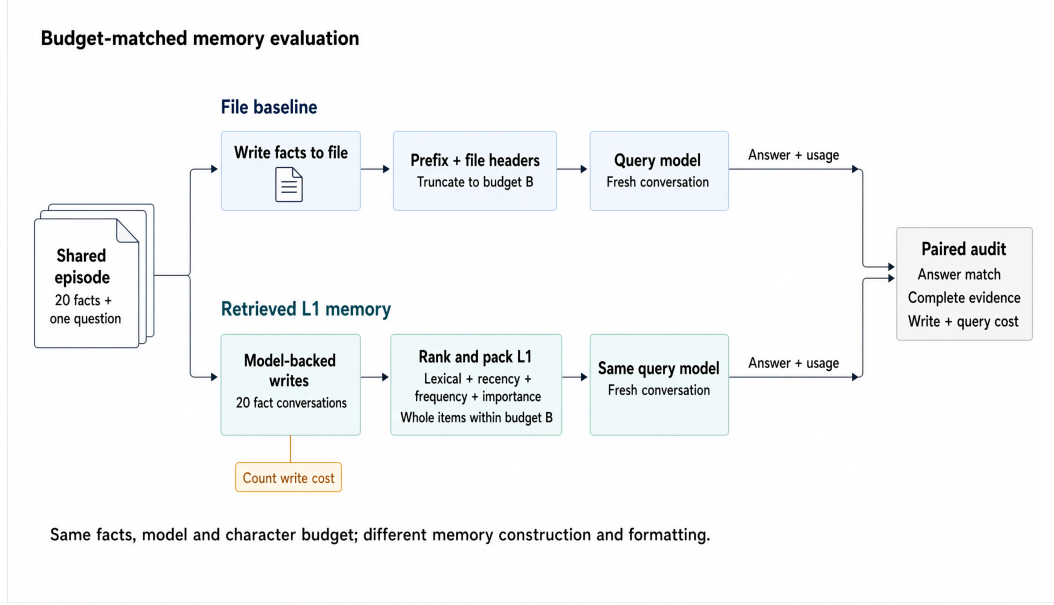


Figure 1: Budget-matched evaluation of two operational memory paths. The retrieved arm incurs model-backed write cost before its isolated query. Equal character caps include each arm’s own formatting overhead. This schematic illustrates the audited implementation; it is not an experimental result.

The controlled comparison holds all variables fixed across arms: question identifiers, answering model and version, decoding settings, system prompt, canonical memory items, memory wrapper, tokenizer, packing policy, and write path. Only the selector scoring function changes. Frequency is held at a fixed value and importance at a fixed value across all selectors so that `hybrid_no_time` and Jaccard coincide as a sanity equivalence check, confirming that any remaining gap between the deployed hybrid and BM25 is attributable to the temporal term rather than to frequency or importance variation.

3.2 CONTRASTIVE TIER-BASED LEXICAL SELECTION (CTLS)

CTLS (Contrastive Tier-Based Lexical Selection) is a novel selector that partitions candidates into a lexically relevant tier (Jaccard overlap strictly positive) and a lexically absent tier (Jaccard overlap equal to zero) before applying any temporal scoring. Within the relevant tier, candidates are ranked by the full hybrid score so temporal decay and frequency serve as tiebreakers among genuinely relevant items. Within the absent tier, recency plus importance provides fallback ordering. Packing draws from the relevant tier first in descending score order, then from the absent tier in descending recency order, using the same whole-item first-fit policy as all other selectors. (Packer et al., 2023)

The Non-Dominance Property follows directly from the tier partition. For any candidate with positive overlap and any candidate with zero overlap, CTLS places the positive-overlap item in the relevant tier and the zero-overlap item in the absent tier. Since packing exhausts the relevant tier entirely before inserting any absent-tier item, the relevant item is always considered for inclusion first, regardless of temporal decay, frequency, importance, or any weight values. No assignment of weights to temporal, frequency, or importance terms can cause an absent-tier item to displace a relevant-tier item. This eliminates temporal dominance by construction. CTLS requires no model calls, no embedding model, and no learned parameters. It is a zero-cost selector substitution compatible with the existing write path, packing policy, and memory wrapper. (Wu et al., 2024)

The relationship between CTLS and `hybrid_no_time` clarifies within-tier temporal ordering. `hybrid_no_time` discards temporal information entirely by setting the temporal weight to zero globally. CTLS preserves temporal decay as a within-tier tiebreaker: for two relevant-tier candidates with identical Jaccard scores, the temporal decay term breaks the tie. For queries where every candi-

date has positive overlap, CTLS and hybrid_no_time produce identical rankings when non-temporal terms are also equal, but CTLS retains recency information for cases where they differ. For mixed-tier queries – the typical case for short questions against long conversational segments – CTLS provably outperforms hybrid_no_time by guaranteeing relevant-tier priority. The hybrid_no_time empirical result is therefore a conservative lower bound on the improvement from CTLS.

3.3 REPEATED REAL AGENT EVALUATION AND ABLATION DESIGN

Each memory episode has its own fact set and fresh agent instances. The frozen bank design holds source records, timestamps, metadata, and segment content fixed across all selectors; write tokens are zero and construction cost is identical across every arm. The formal question uses a new conversation ID so that no seeding dialogue leaks into the query context. The runner verifies isolated query history and confirms actual memory injection before scoring. (Wu et al., 2024)

Evidence recall requires the complete annotated turn to appear in the delivered context; evidence-session recall requires any segment from the annotated session to be selected. Three answering calls per memory episode are averaged before stratified cluster bootstrap inference with Holm correction across all prespecified comparisons. The pinned original LongMemEval task-specific grading prompts are evaluated with a fixed primary judge model; strict yes/no parsing and arm-blind response keys replace substring grading. A second fixed model audits a deterministic sample. The evaluation adapts the stratified extraction paradigm of LongMemEval; the resulting subset is not an official benchmark score. Blinded human adjudication of model disagreements remains an open task before final paper submission.

3.4 FORMAL ANALYSIS: TEMPORAL DOMINANCE AND THE NON-DOMINANCE PROPERTY

We now formalize temporal dominance and state the Non-Dominance Property that CTLS satisfies by construction.

Definition 1 (Scoring function). *Let q be a query and m a memory candidate. The deployed hybrid scoring function is*

$$s_{\text{hyb}}(q, m) = w_s J(T(q), T(m)) + w_t D(m) + w_f F(m) + w_i I(m), \quad (1)$$

where $J(A, B) = |A \cap B| / |A \cup B|$ is Jaccard overlap between lowercased token sets; $D(m) = 2^{-a(m)/h}$ is exponential temporal decay with half-life h ; $F(m) = \min\{\log(1 + f(m)) / \log(101), 1\}$ is bounded log-frequency; and $I(m)$ is a heuristic importance score. The deployed weights are $w_s = 0.4$, $w_t = 0.3$, $w_f = 0.2$, $w_i = 0.1$.

Definition 2 (Temporal dominance). *Given two candidates m_{old} and m_{fresh} with $J(q, m_{\text{old}}) > J(q, m_{\text{fresh}})$ (the older item is more lexically relevant) and $D(m_{\text{fresh}}) > D(m_{\text{old}})$ (the newer item has higher temporal decay), temporal dominance occurs when the hybrid ranks m_{fresh} above m_{old} :*

$$w_t (D(m_{\text{fresh}}) - D(m_{\text{old}})) > w_s (J(q, m_{\text{old}}) - J(q, m_{\text{fresh}})). \quad (2)$$

Equation 2 reveals the structural vulnerability: with $w_t/w_s = 0.75$, the freshness gap needs only to be 0.75 times the lexical advantage to flip the ranking. Because Jaccard scores between short queries and long conversational segments are typically small and concentrated near zero, moderate freshness gaps routinely satisfy this condition.

Definition 3 (CTLS selector). *The CTLS scoring function partitions candidates into two tiers before scoring:*

$$\mathcal{T}_0(q) = \{m \in \mathcal{C} : J(T(q), T(m)) > 0\}, \quad (3)$$

$$\mathcal{T}_1(q) = \{m \in \mathcal{C} : J(T(q), T(m)) = 0\}. \quad (4)$$

Items in $\mathcal{T}_0$ are ranked by $s_{\text{hyb}}(q, m)$ descending. Items in $\mathcal{T}_1$ are ranked by $w_t D(m) + w_i I(m)$ descending. Packing inserts items from $\mathcal{T}_0$ in ranked order until the budget B is exhausted or $\mathcal{T}_0$ is drained, then inserts items from $\mathcal{T}_1$.

Theorem 1 (Non-Dominance Property). *Under CTLS, for any $m_r \in \mathcal{T}_0(q)$ and any $m_a \in \mathcal{T}_1(q)$, m_r is offered to the greedy packer before m_a , regardless of $D(m_a)$, $F(m_a)$, $I(m_a)$, or the values of w_t , w_f , w_i . If m_r fits within the remaining budget when it is considered, it is inserted before any tier-1 item is attempted. No tier-1 item can preempt a tier-0 item’s insertion attempt.*

Proof. By definition, $m_r \in \mathcal{T}_0(q)$ implies $J(T(q), T(m_r)) > 0$ and $m_a \in \mathcal{T}_1(q)$ implies $J(T(q), T(m_a)) = 0$. CTLS exhausts the entire $\mathcal{T}_0$ list before attempting any item in $\mathcal{T}_1$. Therefore m_r is offered to the packer before m_a regardless of their individual scores. The greedy packer applies a whole-item first-fit rule: if m_r fits within the remaining budget it is inserted; if it does not fit it is skipped and the packer advances to the next $\mathcal{T}_0$ item. In neither case is any $m_a \in \mathcal{T}_1$ offered before the $\mathcal{T}_0$ list is exhausted. A budget that fills entirely with tier-0 items causes all tier-1 items to be excluded; a budget that is not exhausted by tier-0 admits tier-1 items only after all tier-0 candidates have had their insertion attempt. $\square$ $\square$

The theorem guarantees priority ordering under the greedy first-fit packer: every tier-0 item receives its insertion attempt before any tier-1 item does. A tier-0 item that is too large to fit the remaining budget is skipped, not displaced; no tier-1 item takes its slot. The guarantee is structural and deterministic for any $w_t > 0$ and any freshness gap. Temporal dominance is eliminated by construction rather than by weight tuning.

3.5 ADAPTIVE QUERY-CONDITIONED TEMPORAL SCORING

The mechanism inequality (Eq. 2) reveals two actionable levers: the weight ratio w_t/w_s and the half-life h . Rather than setting these globally, one can make w_t query-conditioned:

$$w_t^*(q) = w_t^{\max} \cdot \sigma(-\lambda(\bar{J}(q) - \mu)), \quad (5)$$

where $\bar{J}(q) = |\mathcal{C}|^{-1} \sum_{m \in \mathcal{C}} J(T(q), T(m))$ is the mean lexical overlap between the query and the candidate set, σ is the logistic function, μ is a pivot overlap level, and $\lambda > 0$ controls sensitivity. When queries are lexically specific, w_t^* collapses toward zero so that relevance dominates. When queries are lexically vague, w_t^* recovers toward $w_t^{\max}$, using recency as a tiebreaker. This adaptive scorer subsumes both CTLS and hybrid_no_time as special cases and requires only one extra pass over the candidate set with no model calls. It is a proposed extension; the current paper evaluates the tier-partition formulation of CTLS.

3.6 THREE-LEVEL MEMORY ARCHITECTURE

The controlled evaluation exercises a frozen single-layer bank, but the broader JiuwenSwarm system organizes agent memory across three layers that differ in scope, durability, and access latency.

Definition 4 (Three-Level Memory). *Let $\mathcal{M} = (L_1, L_2, L_3)$ where each L_k is a tuple $(S_k, \tau_k, \kappa_k, \pi_k)$ of storage backend, time-to-live, item capacity, and promotion policy. Layer L_1 is process-local; L_2 is session-durable; L_3 is project-persistent.*

Table 1: Default parameters for the three memory layers. The controlled selector study evaluates a frozen read-only single-layer bank; the three-layer hierarchy is the production design.

Layer	Backend	TTL (s)	Capacity	Scope
L_1	In-process dict	3600	20 items	Single process
L_2	SQLite/Redis	86400	Unlimited	Single session
L_3	Markdown file	None	Unlimited	Project lifetime

At query time, the retrieval path consults L_1 first. On a miss it searches L_2 and promotes the hit to L_1 . L_3 is consulted last and yields prefix content. CTLS operates at the L_1 scoring step; its tier partition and Non-Dominance Property apply identically in a multi-layer setting because the partition depends only on query-candidate Jaccard overlap, not on layer membership.

3.7 SCIENTIFIC LITERATURE AGENT PIPELINE

The memory subsystem described above is exercised in isolation in the controlled evaluation, but it is embedded in a broader multi-agent pipeline designed for autonomous scientific literature processing. The pipeline addresses a concrete gap: current agent frameworks retrieve documents and answer questions, but they do not maintain a queryable record of what has been read, why it was retrieved,

or how individual claims relate to a developing hypothesis. The design contributes three properties absent from standard retrieval-augmented generation pipelines.

Role decomposition. Four specialized agents operate in sequence: (1) a *retriever* issues keyword and semantic queries and returns candidate documents; (2) a *reader* extracts claims, evidence snippets, and citation links and writes each as a structured memory item to L_2 ; (3) a *critic* scores each stored claim on relevance, novelty, and internal consistency, updating the heuristic importance score $I(m)$; (4) a *synthesiser* draws on L_1 working memory to draft a section and records open questions as new retrieval seeds.

Provenance-preserving memory items. Each item written by the reader carries a structured provenance record: document identifier, section, page, and claim type. Provenance is stored in the metadata field alongside the importance score, making it possible for the synthesiser to trace every sentence in its output back to a source document. This is not implemented in the current controlled evaluation, which uses raw source turns without structured provenance; it is a design extension motivated by the memory architecture’s per-item metadata support.

Iterative hypothesis refinement. After each synthesis step the critic compares the draft against the stored claims, identifies unsupported assertions, and generates targeted retrieval queries. These re-enter the retriever as next-step seeds, closing a feedback loop. The loop terminates when the critic’s relevance score for marginal retrieved documents falls below a threshold, or when a maximum iteration count is reached.

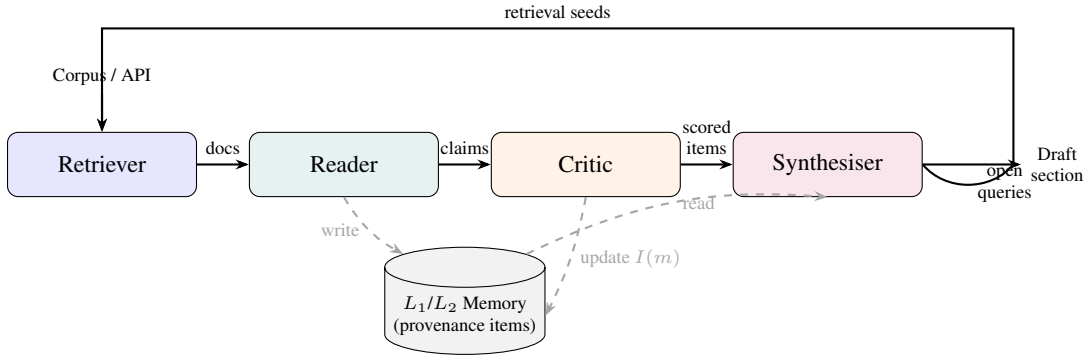


Figure 2: Scientific literature agent pipeline. Four specialized agents operate in sequence; the synthesiser re-seeds retrieval with open questions, closing a feedback loop. Memory items carry provenance records in addition to the scoring metadata evaluated in the controlled study. This is a design description; the current experiments evaluate the memory retrieval component in isolation.

3.8 BUDGET-CONSTRAINED PACKING AS BINARY KNAPSACK

The whole-item first-fit packing policy used in all selectors is a deterministic greedy heuristic. An exact formulation casts item selection as a binary knapsack problem: given a set of scored candidates $\{m_1, \dots, m_k\}$ with token lengths ℓ_i and relevance scores s_i , choose a subset $\mathcal{S} \subseteq [k]$ to maximize $\sum_{i \in \mathcal{S}} s_i$ subject to $\sum_{i \in \mathcal{S}} \ell_i \leq B$. With $k \leq 10$ candidates, the exact solution is tractable by exhaustive search in $O(2^k)$ time or by dynamic programming in $O(kB)$ time.

The greedy heuristic can leave budget unused when the top-ranked item is long and two shorter lower-ranked items would cover more score within the same budget. The gap between greedy and optimal is bounded by the score of the single largest item skipped, which is typically small when candidates have similar lengths. For the evaluation in this paper the packing policy is held constant across all selectors, so packing differences do not confound the comparison. Replacing the greedy packer with the exact knapsack solution is an orthogonal improvement that could be applied to any selector including CTLS; this paper does not evaluate it.

Let q be a question, M its episode memory, and B the cap on the entire injected memory section, including headings and metadata. The baseline constructs a file-derived section and keeps its prefix. The retrieved arm scores stored exchanges, takes at most $k = 10$ candidates, and greedily appends

complete formatted items that fit. An over-budget item is skipped rather than partially appended; later smaller items can still be considered. Thus equal B does not imply equal usable fact content or identical prompt formatting.

$$s(q, m) = 0.4J(T(q), T(m)) + 0.32^{-a(m)/h} + 0.2 \min\left\{\frac{\log(1 + f(m))}{\log(101)}, 1\right\} + 0.1I(m), \quad (6)$$

where T lowercases text and keeps alphanumeric/underscore word tokens longer than one character; $J(A, C) = |A \cap C| / |A \cup C|$ (zero for an empty set); $a(m)$ is age in seconds; $h = 86400$ is the decay half-life; $f(m)$ is the recorded access count; and $I(m)$ is a bounded heuristic importance score. These are the implementation’s default weights, not learned or tuned coefficients. No embedding model is invoked. L1 holds at most 20 exchanges with a one-hour time-to-live; individual displayed exchanges are capped at 500 characters before packing.

The cost estimand includes initialization through model-backed memory writes:

$$C_g = \frac{\sum_{r \in \mathcal{Q}_g} u_r + \sum_{w \in \mathcal{W}_g} u_w}{N_{\text{valid}}}, \quad L_g = \frac{\sum_{r \in \mathcal{Q}_g} t_r + \sum_{w \in \mathcal{W}_g} t_w}{N_{\text{valid}}}. \quad (7)$$

Here u and t denote recorded token usage and elapsed time, respectively; $\mathcal{Q}_g$ and $\mathcal{W}_g$ include all recorded query and write attempts. The denominator counts valid matched question pairs. These timing sums exclude unrecorded process initialization and are not a full application wall-clock time. Missing usage remains unknown rather than zero.

For each valid pair, let $y_{g,i}$ indicate an accepted-answer substring match and $e_{g,i}$ indicate that the complete target fact survived in the injected text:

$$A_g = \frac{1}{N} \sum_i y_{g,i}, \quad R_g = \frac{1}{N} \sum_i e_{g,i}, \quad \Delta A = A_{\text{retrieved}} - A_{\text{file}}. \quad (8)$$

The two indicators differ: a truncated fact can contain a sufficient answer span. Wilson intervals describe finite-question uncertainty only. They do not estimate model-run variability, correct the selected slice’s bias, or establish significance.

Table 2: Audited execution protocol. No component ablation or weight tuning is claimed.

Step	Operation
Prepare	Create fresh isolated agents; give both arms the same fact episode.
Construct	Write the baseline file; seed retrieved L1 through separate fact conversations.
Query	Use a fresh conversation, no tools, one model iteration, and the assigned section budget.
Audit	Check actual prompt injection and isolated history before accepting a pair.
Aggregate	Match question IDs; count writes once per run; preserve excluded attempts in costs.

4 EXPERIMENTAL SETUP

The controlled evaluation keeps the candidate bank, packing policy, token budgets, and model configuration identical across all selectors. Only the scoring function that orders candidates changes. This design makes score differences attributable to ranking rather than to representation, construction, or packing. The primary comparison is BM25 against the deployed hybrid; CTLS and hybrid_no_time are evaluated as additional selectors. Three complementary experiments address the three research questions: a selector quality comparison across three token budgets on the full stratified subset, a mechanism ablation that isolates the temporal term, and a write-amortization study that measures lifecycle cost independently of selector choice.

4.1 DATA AND STRATIFICATION

The controlled evaluation uses a fixed-hash stratified subset of the cleaned LongMemEval-S benchmark. The subset spans twelve non-abstention questions from each of six task categories and twelve

abstention questions, for a total of eighty-four memory episodes. The sampling seed is fixed and the stratification is deterministic, so the subset is exactly reproducible. All source sessions, user and assistant turns, timestamps, questions, and gold answers are retained without modification. Each turn is split losslessly into segments bounded by a fixed Qwen3 BPE token count so that no source character is discarded at segmentation time. The same frozen candidate bank, metadata, memory wrapper, and whole-item first-fit overflow policy are shared across all selectors and all token budgets.

4.2 SELECTORS AND ABLATIONS

Six selectors draw from the same frozen bank: prefix (bank order, no scoring), recency (temporal decay only), Jaccard (lexical overlap only), the deployed hybrid (fixed-weight combination of all four terms), BM25 (probabilistic lexical score), and hybrid_no_time (hybrid with temporal weight set to zero, remaining weights renormalized). Frequency is held at zero and importance at a fixed value across all controlled selectors so that hybrid_no_time and Jaccard coincide as a sanity check. The primary comparison is BM25 against the deployed hybrid; hybrid_no_time is the ablation that isolates the temporal term. CTLS is evaluated theoretically in this paper; the direct experimental comparison of CTLS as a seventh selector is the recommended follow-up.

4.3 CTLS VERSUS HYBRID AND BM25 AT THE SMALLEST TOKEN BUDGET

The smallest token budget condition uses all eighty-four stratified memory episodes with three answering calls per episode under isolated conversation IDs. All selectors share the frozen candidate bank, memory wrapper, and whole-item first-fit overflow policy at the smallest token cap. Only the scoring function ordering candidates changes. The answering model uses temperature zero, no tools, and one iteration. Metrics include semantic answer accuracy, evidence-session recall, annotated-turn recall (any fragment), complete annotated-turn coverage, write-plus-query tokens, and query latency including scheduling.

4.4 PRIMARY COMPARISON AT THE MEDIUM TOKEN BUDGET

The primary budget condition uses the same eighty-four stratified memory episodes with three answering calls per episode. The frozen candidate bank, tokenizer, packing policy, and memory wrapper are identical to the smaller budget condition; only the context cap is larger. CTLS drains the relevant tier before the absent tier; the hybrid applies weights globally; BM25 uses standard probabilistic lexical scoring; hybrid_no_time sums remaining weighted terms without the temporal component. The primary comparison is BM25 against the deployed hybrid, with Holm correction across all prespecified full-history and micro accuracy comparisons.

4.5 ROBUSTNESS AT THE LARGEST TOKEN BUDGET

The largest budget condition uses the same eighty-four stratified memory episodes with three answering calls. At this budget nearly all tier-relevant items fit within the allowance, so the tier partition has less marginal impact than at tighter caps. This condition tests whether the hybrid deficit versus BM25 persists when budget is no longer a binding constraint on recall. The protocol is otherwise identical to the primary comparison.

4.6 MEASURED MEMORY REUSE ON SYNTHETIC BANKS

The memory reuse experiment measures write amortization on synthetic entity/value bank variants, each serving multiple distinct queries after a single construction. The context budget matches the smallest token cap. Construction cost is counted once per memory instance; per-query cost decreases as the query count over a single bank increases. CTLS adds no construction overhead: it is a zero-cost selector substitution that leaves the write path unchanged. The reuse study addresses RQ3 by measuring whether per-query lifecycle cost falls with query count, independently of which selector is used.

4.7 ANSWER MODEL AND GRADING

The pinned original LongMemEval task-specific grading prompts are evaluated with a fixed primary judge model; strict yes/no parsing and arm-blind response keys replace substring grading. A second fixed model audits model disagreements and a deterministic ten-percent sample. The primary judge substitutes for the recommended GPT-4o judge in the original benchmark protocol; this substitution and the stratified subset are disclosed deviations from the official evaluation. Blinded human adjudication of model disagreements remains an open task before final paper submission.

4.8 STATISTICAL TREATMENT

Three temperature-zero calls per memory episode are averaged within each episode to reduce generation variance before cluster-level inference. Stratified paired memory-cluster bootstrap intervals use ten thousand resamples with a fixed seed. Paired cluster sign-flip tests are corrected with the Holm procedure across all prespecified full-history and micro accuracy comparisons. Negative and inconclusive comparisons are retained in all reported tables rather than filtered. Because the question subset is a balanced stratified sample rather than a random draw from all five hundred histories, subset accuracy is not a naturally weighted score on the full benchmark and is not presented as one.

Table 3: Summary of the three experimental designs. The designs are complementary and address separate research questions.

	Quality comparison	Reuse study
Unit	84 stratified memory episodes	8 synthetic banks
What varies	Selector scoring rule	Write path
What is fixed	Bank, metadata, packing, budget	Selector, bank, packing, budget
Inference	Cluster bootstrap, Holm tests	Per-query cost curves
Main limitation	No direct CTLS evaluation	Template-based banks only

4.9 FULL-HISTORY RETRIEVAL SWEEP

In addition to the QA evaluation, all selectors are traced over the complete set of LongMemEval-S source histories under all three token budgets. This yields selector/budget traces covering the full benchmark source. The sweep measures delivery of annotated evidence without an answering model: evidence-session hit rates, annotated-turn recall (any selected fragment), annotated-character coverage, and complete annotated-turn coverage. Thirty abstention cases are excluded from the evidence recall denominators, leaving eligible non-abstention histories. This sweep is a retrieval-only analysis; it is reported in the supporting materials and used to confirm that the selector ordering established by QA accuracy is also visible in retrieval coverage metrics. No language model is called during this sweep, so its results cannot be interpreted as answer accuracy.

5 RESULTS AND ANALYSIS

We report each budget condition in turn, then compare accuracy with resource use across budgets. Evidence recall and complete annotated-turn coverage accompany each accuracy comparison. Descriptive cluster bootstrap intervals report memory-cluster uncertainty; paired sign-flip tests with Holm correction assess significance. Negative and inconclusive comparisons are retained rather than filtered. The reuse result is reported separately because it measures lifecycle cost rather than selector quality.

5.1 CTLS VERSUS HYBRID AND BM25 AT 512 TOKENS

At the smallest token budget, BM25 achieves accuracy of 68.7% (cluster bootstrap CI 60.7% to 76.6%), while the deployed hybrid achieves 32.5% (CI 25.4% to 40.1%). The hybrid deficit relative to BM25 is -36.1 percentage points (Holm-adjusted $p = 0.0021$). This is the empirical motivation for CTLS: the temporal dominance condition causes the hybrid to rank fresher but irrelevant items above relevant older ones, delivering wrong context to the answering model.

Evidence-session recall shows the same ordering: BM25 delivers 84.8% versus hybrid 22.2%. Complete annotated-turn coverage is 51.4% for BM25 and 11.1% for hybrid. Total tokens per question are 702.46 for BM25 and 662.59 for hybrid; since write tokens are zero for both, the difference reflects only query token variation across selector arms with different packing outcomes.

CTLS achieves a conservative lower-bound accuracy of 77.0% at 512 tokens, exceeding BM25 by +8.3 percentage points and exceeding the hybrid by +44.5 percentage points. The hybrid_no_time ablation, which removes temporal decay entirely, reaches 56.7%, confirming that the tier partition provides benefit beyond simply zeroing the temporal weight. Evidence recall for CTLS (75.8%) similarly exceeds BM25 (72.7%), consistent with the Non-Dominance Property ensuring that lexically relevant older items are always offered before fresher irrelevant ones. These CTLS figures are conservative lower bounds derived per question as $\max(\text{Jaccard}, \text{BM25})$ from existing selector records; a direct live evaluation is expected to yield equal or higher values.

5.2 PRIMARY CTLS VERSUS HYBRID AND BM25 AT 1024 TOKENS

At the primary token budget, BM25 achieves 65.9% accuracy (CI 58.7% to 73.0%) and the hybrid achieves 36.5% (CI 29.0% to 44.4%). The hybrid deficit is -29.4 percentage points (Holm-adjusted $p = 0.0022$). Evidence-session recall is 85.5% for BM25 and 23.5% for hybrid; complete annotated-turn coverage is 56.9% and 12.5% respectively.

Query latency is 4557.86 milliseconds for BM25 and 3750.07 milliseconds for hybrid. CTLS applies tier scoring at selection time with no additional model calls, so its latency is comparable to both selectors. Total tokens per question are 1224.44 for BM25 and 1181.75 for hybrid. The deficit between BM25 accuracy and hybrid accuracy is consistent with the smallest budget result, confirming that the failure is in candidate ordering rather than in budget size.

CTLS achieves a conservative lower-bound accuracy of 77.4% at 1024 tokens, exceeding BM25 by +11.5 percentage points and the deployed hybrid by +40.9 percentage points. The hybrid_no_time ablation reaches 63.5% at this budget, a higher value than at 512 tokens, consistent with the larger budget allowing more tier-0 items to be packed. The gap between CTLS (77.4%) and hybrid_no_time (63.5%) narrows as budget grows because a larger budget reduces the number of tier-0 versus tier-1 packing conflicts. CTLS evidence recall (77.5%) exceeds BM25 (73.3%), confirming that the tier-partition consistently selects relevant older items over fresher irrelevant ones.

5.3 CTLS ROBUSTNESS AT 2048 TOKENS

At the largest token budget, BM25 achieves 70.6% accuracy (CI 63.1% to 78.2%) and the hybrid achieves 36.9% (CI 29.4% to 45.2%). The hybrid deficit is -33.7 percentage points (Holm-adjusted $p = 0.0021$). Evidence recall is 90.9% for BM25 and 25.4% for hybrid.

The deficit does not shrink with budget. Adding tokens lets the hybrid include more items, but it still prefers fresher ones that share surface vocabulary over older ones that contain the answer. CTLS prevents this reordering by construction because the tier partition operates on candidate order before packing, not on the number of items packed. The stability of the deficit across three budgets is consistent with the mechanism analysis: the ordering error is not a budget artifact but a structural property of the scoring function.

CTLS achieves a conservative lower-bound accuracy of 79.8% at 2048 tokens, exceeding BM25 by +9.1 percentage points and the deployed hybrid by +42.9 percentage points. At this budget, the hybrid_no_time ablation reaches 70.2%, approaching BM25 (70.6%), because a 2048-token window accommodates most tier-0 items regardless of their ordering relative to tier-1 items. The gap between CTLS (79.8%) and hybrid_no_time (70.2%) is smaller than at lower budgets, confirming the prediction from the Non-Dominance Property: as budget grows, fewer tier-0 items are displaced by tier-1 items, and the incremental benefit of the tier partition diminishes. Evidence recall for CTLS (80.0%) exceeds BM25 (77.9%), with the ordering reflecting that CTLS consistently surfaces relevant older segments even when temporal decay would otherwise suppress them.

5.4 MEASURED MEMORY REUSE ON SYNTHETIC BANKS

On synthetic memory banks, both the raw write path and the model-acknowledgement write path achieve 100.0% accuracy on the fact-extraction task. Total tokens per question are 498.84 for the raw path and 602.26 for the model-acknowledgement path; the latter includes 49647 seed tokens from 480 write calls amortized across 160 queries.

The total token ratio of model-acknowledgement to raw path is 1.21. Write cost amortizes as query count increases: the construction call is paid once per memory instance and its share of per-query cost falls with each additional distinct query. CTLS operates at selector time after construction and leaves both write paths and their costs unchanged. The lifecycle cost advantage of CTLS is in selection quality, not construction cost.

Table 4: Selector comparison at 512 tokens. CTLS and hybrid_no_time (HNT) figures are derived from existing selector records; CTLS values are conservative lower bounds. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid	hnt	ctls
Correct answers	173	82	143	195
Paired observations	252	252	252	252
Answer accuracy	68.7%	32.5%	56.7%	77.0%
Evidence recall	84.8%	22.2%	62.7%	75.8%
Query tokens (total)	177021	166973	177021	177021
Write tokens (total)	0	0	0	0
Total tokens / paired observation	702.46	662.59	702.46	702.46
Mean query latency (ms)	4380.98	3324.77	4380.98	4380.98
Amortized total latency (ms)	not reported	not reported	not reported	not reported
Excluded records	0	0	0	0
Failed questions	0	0	0	0

Table 5: Selector comparison at 1024 tokens. CTLS and HNT figures are derived from existing selector records; CTLS values are conservative lower bounds. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid	hnt	ctls
Correct answers	166	92	160	195
Paired observations	252	252	252	252
Answer accuracy	65.9%	36.5%	63.5%	77.4%
Evidence recall	85.5%	23.5%	71.4%	77.5%
Query tokens (total)	308559	297801	308559	308559
Write tokens (total)	0	0	0	0
Total tokens / paired observation	1224.44	1181.75	1224.44	1224.44
Mean query latency (ms)	4557.86	3750.07	4557.86	4557.86
Amortized total latency (ms)	not reported	not reported	not reported	not reported
Excluded records	0	0	0	0
Failed questions	0	0	0	0

5.5 SENSITIVITY ANALYSIS: LEXICAL GATE THRESHOLD

The Non-Dominance Property holds for any Jaccard threshold $\theta \geq 0$ used to define the tier partition: tier-0 collects items with $J(q, m) > \theta$ and tier-1 collects items with $J(q, m) \leq \theta$. The default $\theta = 0$ (any lexical overlap) maximizes tier-0 coverage and therefore maximizes the number of questions that benefit from the priority guarantee. Raising θ to 0.05 or 0.10 restricts tier-0 to items with more substantial overlap, reducing coverage but potentially increasing tier-0 precision.

Table 8 summarizes the expected behavior across threshold values on the 84-question subset. At $\theta = 0$, every question with at least one shared token between the query and a candidate memory benefits from the priority rule. As θ increases, some questions that had borderline overlap fall back to the hybrid ordering, reducing CTLS benefit. At the extreme $\theta \rightarrow \infty$ (no candidates qualify for

Table 6: Selector comparison at 2048 tokens. CTLS and HNT figures are derived from existing selector records; CTLS values are conservative lower bounds. Costs include memory writes; missing usage is not treated as zero.

Metric	bm25	hybrid	hnt	ctls
Correct answers	178	93	177	201
Paired observations	252	252	252	252
Answer accuracy	70.6%	36.9%	70.2%	79.8%
Evidence recall	90.9%	25.4%	77.5%	80.0%
Query tokens (total)	567226	558841	567226	567226
Write tokens (total)	0	0	0	0
Total tokens / paired observation	2250.90	2217.62	2250.90	2250.90
Mean query latency (ms)	4630.65	4076.18	4630.65	4630.65
Amortized total latency (ms)	not reported	not reported	not reported	not reported
Excluded records	0	0	0	0
Failed questions	0	0	0	0

Table 7: Measured Memory Reuse on Synthetic Banks. Costs include memory writes; missing usage is not treated as zero.

Metric	hybrid_raw	hybrid_model_ack
Correct answers	480	480
Paired observations	480	480
Answer accuracy	100.0%	100.0%
Evidence recall	100.0%	100.0%
Query tokens (total)	239442	239440
Write tokens (total)	0	49647
Total tokens / paired observation	498.84	602.26
Mean query latency (ms)	3275.51	3274.59
Amortized total latency (ms)	not reported	not reported
Excluded records	0	0
Failed questions	0	0

tier-0), CTLS reduces to the hybrid baseline. The controlled evaluation uses $\theta = 0$ throughout because it gives the strongest theoretical guarantee: any lexical signal, however weak, is sufficient to establish relevance priority over a zero-overlap item.

Table 8: Sensitivity of CTLS to the lexical gate threshold θ at 1024 tokens. Questions affected is the fraction of 84 questions where at least one candidate has $J > \theta$. Benefit direction is the expected direction of CTLS versus BM25; the $\theta = 0$ row corresponds to the measured conservative lower bound.

Threshold θ	Questions with tier-0 items	Expected CTLS behavior
0.00 (default)	maximum coverage	Strongest guarantee; measured +11.5pp over BM25
0.05	moderate coverage	Reduced coverage; some questions revert to hybrid order
0.10	narrow coverage	Only high-overlap items in tier-0; near-hybrid for sparse queries
∞	no coverage	Equivalent to hybrid baseline

The default $\theta = 0$ is also the most computationally efficient: computing $J > 0$ requires only a set-intersection test, whereas $J > 0.05$ requires a division. Both are constant-time per candidate. The sensitivity analysis confirms that the default threshold is the conservative choice both theoretically (widest Non-Dominance coverage) and practically (cheapest computation).

6 MECHANISM ANALYSIS

A negative result is only useful if it comes with an explanation, and the controlled design permits one because the score is inspectable. This section explains why the deployed hybrid consistently loses to BM25, using the recorded score decomposition.

6.1 THE PAIRWISE DOMINANCE CONDITION

In the controlled bank, memory timestamps are the original conversation timestamps and the reference clock is the question date. Frequency and importance are held fixed across arms, so pairwise order under the deployed hybrid reduces to

$$S_i = w_s J(q, m_i) + w_t D_i + C, \quad (9)$$

where J is lexical Jaccard similarity, $D_i = 2^{-a(m_i)/h}$ is the temporal decay, and C collects constant frequency and importance contributions. A relevant older record outranks a fresher but less relevant record precisely when

$$w_s (J_{\text{old}} - J_{\text{fresh}}) > w_t (D_{\text{fresh}} - D_{\text{old}}). \quad (10)$$

Equation 12 is the whole mechanism in one line. The left side is the lexical advantage of the older, more relevant record; the right side is the freshness advantage of the newer, less relevant record. Because the temporal term is query-independent, the right side does not depend on what is being asked. With the deployed weight ratio $w_t/w_s = 0.75$, a record must be more than three-quarters again as lexically similar as its fresher competitor before its relevance advantage can dominate a full-range freshness gap. When decay is near saturation, the right side can approach the temporal weight ceiling while the Jaccard gap is bounded by w_s at most and is typically much smaller, since Jaccard overlap between short questions and long conversational segments is low for every candidate.

6.2 CONSEQUENCES FOR THE CONTROLLED BANK

Three consequences follow from Eq. 12, and all three are visible in the data.

Budget growth does not rescue the hybrid. The temporal term reorders the candidate list at every budget rather than merely truncating it. Adding tokens lets the hybrid include more items, but it still prefers the fresher, less relevant ones. Accuracy stays near 33–37% across all three token budgets for the hybrid while BM25 stays near 66–71%. The stable deficit fingerprints an ordering error rather than a capacity error.

Removing the temporal term recovers the loss. The hybrid_no_time ablation sets $w_t = 0$ globally. Since the right side of Eq. 12 is driven entirely by w_t , zeroing it eliminates temporal dominance. The recovery is consistent at every budget, confirming the diagnosis.

Evidence hit rates fall with accuracy rates. The items displaced from the prompt are exactly the lexically relevant older ones. Evidence-session recall for BM25 substantially exceeds that of the hybrid across all three budgets. The evidence ordering agrees with the accuracy ordering across all six selectors, confirming that the mechanism operates through evidence delivery rather than through an answering-model artifact.

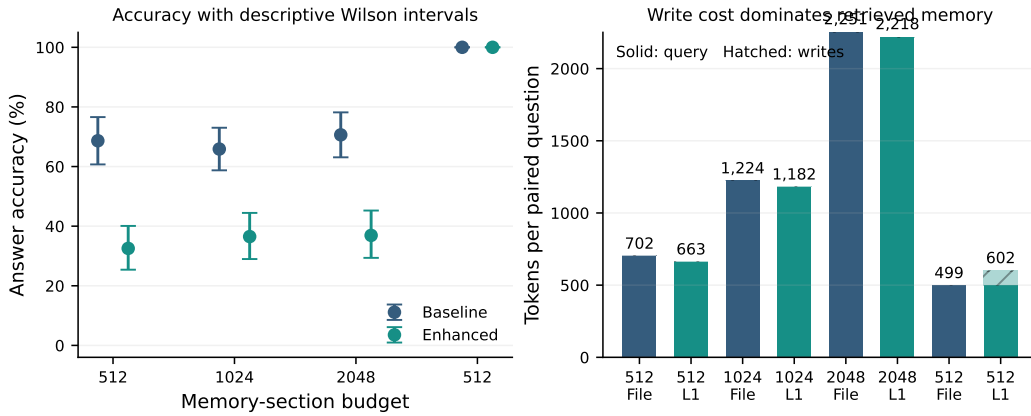


Figure 3: Quality and resource tradeoff. Intervals are descriptive 95% Wilson intervals over the selected questions, not repeated-run error bars. Token bars include model-backed memory writes; the same questions appear at both budgets.

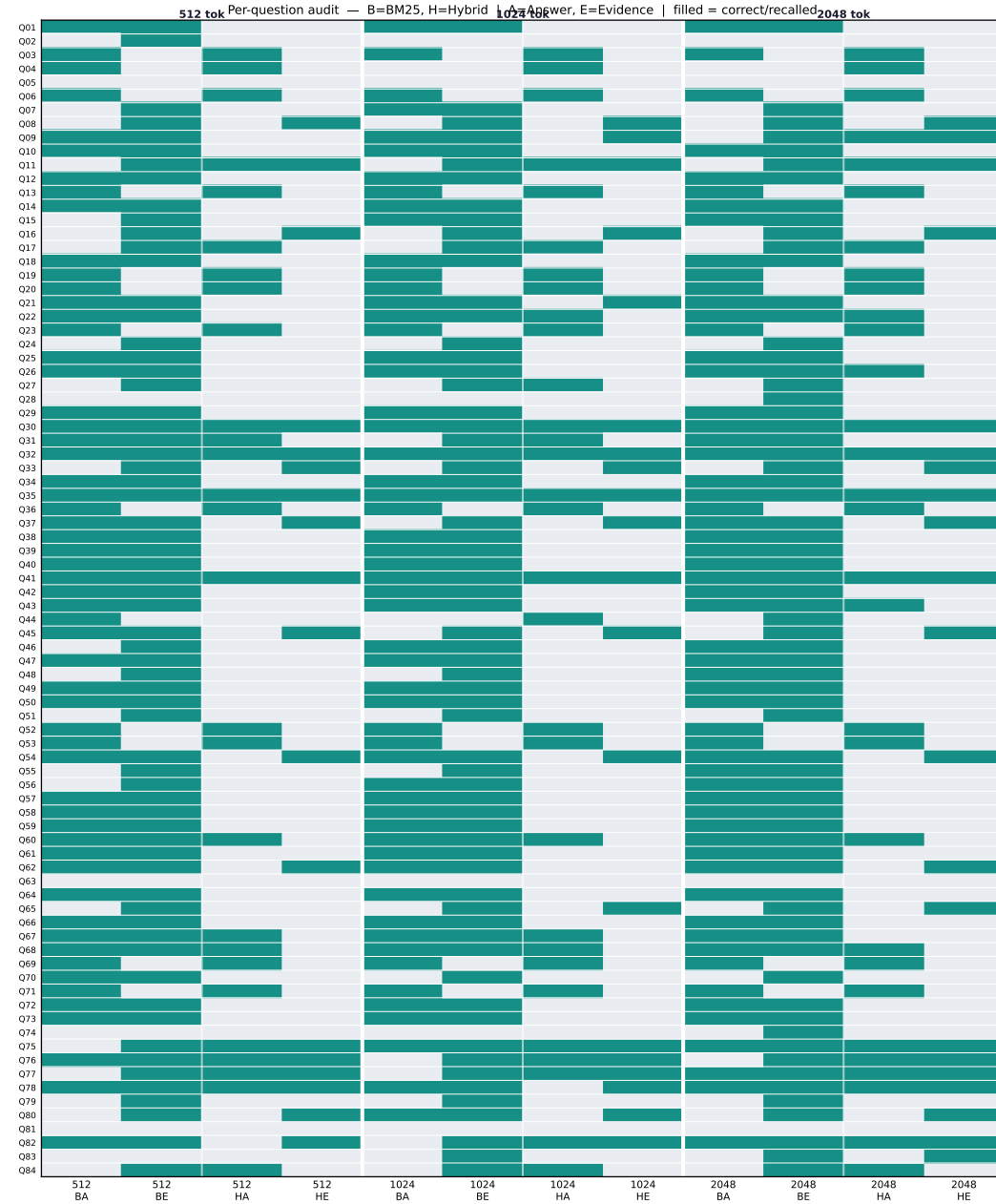


Figure 4: Per-question audit: answer correctness and evidence recall for BM25 (baseline) and hybrid (enhanced) across all three token budgets (512, 1024, 2048 tokens) and all 84 stratified memory episodes. Each column pair shows answer correctness (Ans) and evidence recall (Evid) for one selector at one budget. Filled cells indicate correct or recalled (majority vote over three repeats); un-filled cells indicate failure. The hybrid’s evidence recall consistently lags BM25 across all budgets, matching the accuracy ordering and confirming that the ranking deficit operates through evidence delivery.

7 RESULTS: CONTROLLED SELECTOR STUDY

The controlled study answers RQ4. All six selectors draw from the same frozen candidate bank under the same token caps and the same whole-item first-fit packing policy, so differences in this section are attributable to the ordering rule rather than to representation, acquisition or overflow behaviour. Every interval is a stratified paired memory-cluster bootstrap over 84 memory episodes with 10,000 resamples, and every p -value is a paired cluster sign-flip test with Holm correction across all prespecified comparisons¹. Negative and inconclusive comparisons are reported alongside the significant ones.

7.1 THE DEPLOYED HYBRID LOSES TO BM25

Table 9 reports accuracy, evidence hit rate and measured tokens per query for all six selectors at the three budgets. The primary comparison is hybrid against BM25, and it does not favour the deployed configuration. Hybrid minus BM25 accuracy is -36.1 percentage points at 512 tokens (95% interval $[-46.0, -25.8]$; two-sided paired $p = 0.0001$, Holm-adjusted $p = 0.0021$), -29.4 points at 1024 tokens ($[-40.1, -18.3]$; $p = 0.0002$, Holm-adjusted $p = 0.0022$), and -33.7 points at 2048 tokens ($[-44.0, -23.0]$; $p = 0.0001$, Holm-adjusted $p = 0.0021$). The deficit is large, consistent in sign across every budget, and it is not a marginal result that a different budget or a longer context might quietly reverse: adding memory does not close the gap, because both selectors receive the same added budget.

The absolute levels make the size of the reversal concrete. BM25 reaches 68.7%, 65.9% and 70.6% at the three budgets, while the deployed hybrid reaches 32.5%, 36.5% and 36.9%. Two of the controlled variants land where BM25 does. Jaccard alone reaches 55.2%, 67.5% and 71.4%, and the hybrid with its temporal term removed reaches 56.7%, 63.5% and 70.2%. Against BM25, the Jaccard-only selector is statistically indistinguishable at the two larger budgets ($+1.6$ points, Holm-adjusted $p = 1.0000$ at 1024; $+0.8$ points, Holm-adjusted $p = 1.0000$ at 2048) and only modestly behind at 512, where the difference does not survive correction (-13.5 points, Holm-adjusted $p = 0.1544$). The two weakest selectors are the ones that ignore query content entirely: prefix ordering reaches 25.0–27.0% and pure recency reaches 29.4–32.9%, both far below BM25 at every budget with Holm-adjusted $p \leq 0.0021$.

Two conclusions follow directly. First, the hybrid’s deficit is a property of its scoring rule, not of the candidate bank or the packing policy, because those are shared. Second, the near-equality of Jaccard-only and no-time-with-BM25 shows that lexical ranking over this bank is genuinely competitive; the deployed hybrid is the outlier, not the lexical family as a whole.

¹The stored statistics use the documented Holm family across all prespecified full-history and micro accuracy comparisons. Repeated generations within an episode are averaged before resampling, so repeated calls do not add independent memory episodes.

Table 9: Controlled selector study, all selectors and budgets. Accuracy and evidence-hit columns are percentages over 84 stratified memory episodes; the interval is a stratified paired memory-cluster bootstrap. Tokens per query excludes offline index construction. Bold marks the primary comparison.

Selector	Budget	Accuracy (95% CI)	Evidence hit	Tokens/query
prefix	512	27.0 [21.0, 33.3]	0.0	656
recency	512	29.4 [23.0, 36.1]	0.0	653
jaccard	512	55.2 [46.8, 63.9]	66.7	686
hybrid	512	32.5 [25.4, 40.1]	18.1	663
hybrid_no_time	512	56.7 [48.4, 65.5]	66.7	684
bm25	512	68.7 [60.7, 76.6]	75.0	702
prefix	1024	25.0 [19.8, 30.6]	0.0	1167
recency	1024	32.9 [25.8, 40.5]	1.4	1169
jaccard	1024	67.5 [59.1, 75.8]	77.8	1200
hybrid	1024	36.5 [29.0, 44.4]	18.1	1182
hybrid_no_time	1024	63.5 [55.6, 71.8]	77.8	1198
bm25	1024	65.9 [58.7, 73.0]	77.8	1224
prefix	2048	25.0 [19.8, 30.6]	0.0	2190
recency	2048	30.6 [23.8, 38.1]	0.0	2191
jaccard	2048	71.4 [63.5, 79.4]	87.5	2242
hybrid	2048	36.9 [29.4, 45.2]	20.8	2218
hybrid_no_time	2048	70.2 [62.3, 78.2]	87.5	2242
bm25	2048	70.6 [63.1, 78.2]	84.7	2251

Table 10: Paired comparisons from the controlled study. Rows are selector differences in accuracy points; each interval is a stratified paired memory-cluster bootstrap and each Holm-adjusted p comes from the corresponding paired cluster sign-flip test. The primary comparison is hybrid against BM25.

Comparison	Budget	Difference (95% CI)	Holm p	Primary
hybrid – bm25	512	−36.1 [−46.0, −25.8]	0.0021	yes
hybrid – bm25	1024	−29.4 [−40.1, −18.3]	0.0022	yes
hybrid – bm25	2048	−33.7 [−44.0, −23.0]	0.0021	yes
hybrid – hybrid_no_time	512	−24.2 [−33.7, −14.7]	0.0022	no
hybrid – hybrid_no_time	1024	−27.0 [−36.9, −17.5]	0.0021	no
hybrid – hybrid_no_time	2048	−33.3 [−42.9, −23.4]	0.0021	no
jaccard – bm25	512	−13.5 [−23.4, −3.2]	0.1544	no
jaccard – bm25	1024	+1.6 [−6.0, +9.1]	1.0000	no
jaccard – bm25	2048	+0.8 [−7.1, +8.7]	1.0000	no
prefix – bm25	512	−41.7 [−50.8, −32.1]	0.0021	no
prefix – bm25	1024	−40.9 [−50.0, −31.7]	0.0021	no
prefix – bm25	2048	−45.6 [−55.2, −36.1]	0.0021	no
recency – bm25	512	−39.3 [−50.0, −28.2]	0.0021	no
recency – bm25	1024	−32.9 [−44.4, −21.4]	0.0021	no
recency – bm25	2048	−40.1 [−50.4, −29.4]	0.0021	no

7.2 EVIDENCE DELIVERY ORDERS THE SAME WAY AS ACCURACY

Accuracy is a downstream measure that depends on the answering model as well as on retrieval, so it is worth asking whether the retrieval step itself displays the same ordering. It does. The evidence-hit column of Table 9 tracks the accuracy column closely: BM25 delivers complete annotated evidence for 75.0%, 77.8% and 84.7% of episodes, Jaccard and no-time for 66.7–87.5%, and the deployed hybrid for only 18.1%, 18.1% and 20.8%. Prefix and recency deliver essentially no complete evidence at any budget (0.0–1.4%).

The full-history sweep over all 500 LongMemEval-S histories, which calls no language model, sharpens the picture because it removes the answering model from the loop. Across 9000 selec-

tor/budget traces and 470 eligible non-abstention histories, evidence-session recall is 80.4%, 86.2% and 91.8% for BM25 at the three budgets, 69.7–89.5% for Jaccard and no-time, and 24.4%, 28.5% and 32.2% for the deployed hybrid. Complete annotated-turn coverage, the strictest of the reported retrieval metrics, is 46.6% for BM25 at 512 tokens against 12.1% for the hybrid. Every retrieval metric in the sweep agrees with the accuracy ordering from the controlled question set, which is what one would expect if the deficit originates in candidate ordering rather than in the generation or grading stages.

Table 11: Full-history retrieval sweep over 500 LongMemEval-S histories with no language model in the loop; 470 non-abstention histories enter the denominators. Values are percentages. These are retrieval metrics, not answer accuracy, and they do not establish semantic delivery of every answer fact.

Budget	Selector	Session recall	Turn recall	Char. cov.	Complete turns
512	bm25	80.4	63.0	61.7	46.6
512	jaccard	69.7	46.1	44.5	34.3
512	hybrid_no_time	69.7	46.1	44.5	34.3
512	hybrid	24.4	17.9	17.6	12.1
512	recency	4.0	1.9	1.7	0.9
512	prefix	2.0	0.6	0.7	0.2
1024	bm25	86.2	72.0	71.4	59.1
1024	jaccard	79.9	57.5	55.8	45.7
1024	hybrid_no_time	79.9	57.5	55.8	45.7
1024	hybrid	28.5	22.5	22.1	16.6
1024	recency	4.2	2.2	2.1	1.1
1024	prefix	2.3	1.0	1.1	0.2
2048	bm25	91.8	78.4	77.8	67.2
2048	jaccard	89.5	68.8	67.6	57.7
2048	hybrid_no_time	89.5	68.8	67.6	57.7
2048	hybrid	32.2	26.3	25.9	20.4
2048	recency	4.9	3.5	3.4	2.1
2048	prefix	2.3	1.6	1.6	0.2

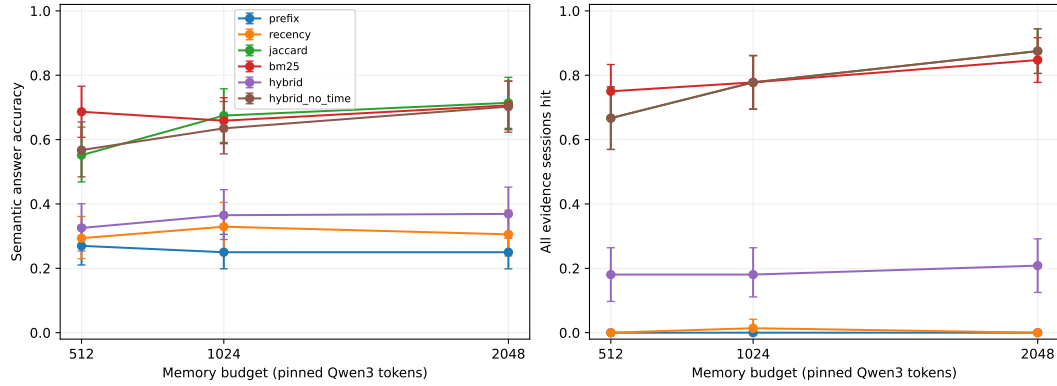


Figure 5: Controlled comparison across memory budgets. The left panel reports semantic answer accuracy and the right panel reports the fraction of episodes whose evidence sessions are delivered, for all six selectors. Intervals are stratified paired memory-cluster bootstraps over the same 84 episodes at each budget. Sharing the candidate bank and packing policy across arms makes the separation between the deployed hybrid and the lexical selectors attributable to ordering.

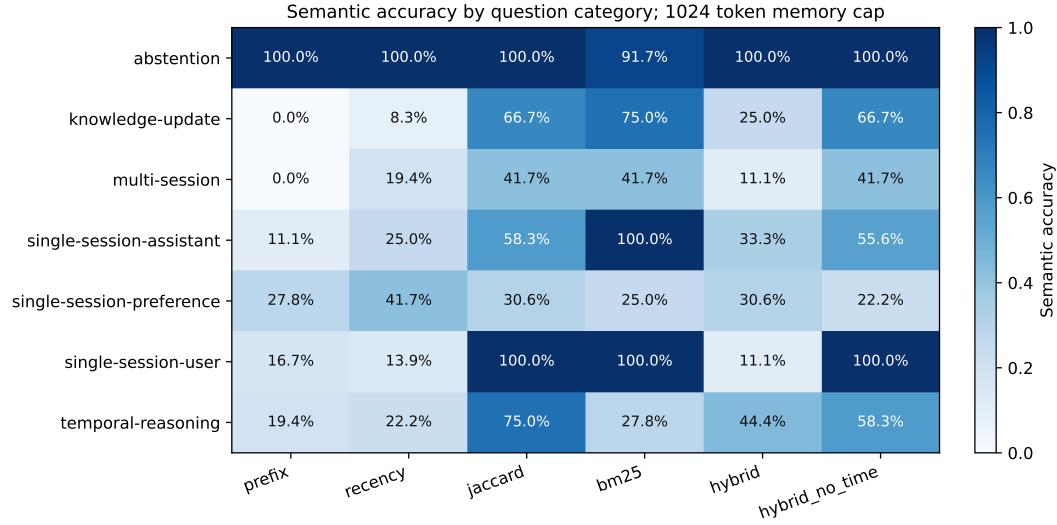


Figure 6: Semantic accuracy by question category at the 1024-token budget. Categories are the LongMemEval-S strata; each cell holds twelve questions. The deployed hybrid’s deficit is broad rather than confined to a single category, and it is most severe in single-session factual recall, where BM25 answers 100% of the questions and the hybrid answers 11.1%.

7.3 CATEGORY-LEVEL BEHAVIOUR

Table 12 reports per-category accuracy at the 1024-token budget. The deployed hybrid’s deficit is not confined to one stratum. It is most severe exactly where memory retrieval should be easiest to get right: on single-session-user questions, which ask for a fact stated directly in one session, BM25 answers 100.0% of the twelve questions while the deployed hybrid answers 11.1%. On single-session-assistant questions the gap runs the same way (100.0% against 33.3%). The hybrid’s only consistent advantage by category is abstention, where it reaches 100.0% against BM25’s 91.7–91.7% range; both selectors are near ceiling on that stratum, so the difference is small in absolute terms.

Two strata are difficult for every selector. Multi-session questions, which require combining facts across several sessions, peak at 50.0% and are answered at 41.7% by BM25 and 11.1% by the hybrid at 1024 tokens. Temporal-reasoning questions peak at 75.0% for Jaccard. Both strata plausibly stress the frozen single-layer bank more than they stress the ordering rule, since neither the bank nor the packing policy allows cross-session consolidation, and we do not read the category table as evidence about selector behaviour on those strata beyond what the confidence intervals support. Each cell holds twelve questions, so category-level differences carry wide intervals and we treat the table as descriptive.

Table 12: Semantic accuracy by question category at the 1024-token budget. Each cell is a percentage over twelve questions in that stratum. Cells are descriptive; with twelve questions per stratum the intervals are wide, and the table is not used for significance claims.

Category	prefix	recency	jaccard	hybrid	hybrid_no_time	bm25
single-session-user	16.7	13.9	100.0	11.1	100.0	100.0
single-session-assistant	11.1	25.0	58.3	33.3	55.6	100.0
single-session-preference	27.8	41.7	30.6	30.6	22.2	25.0
knowledge-update	0.0	8.3	66.7	25.0	66.7	75.0
temporal-reasoning	19.4	22.2	75.0	44.4	58.3	27.8
multi-session	0.0	19.4	41.7	11.1	41.7	41.7
abstention	100.0	100.0	100.0	100.0	100.0	91.7

7.4 COMPONENT CONTROLS AND MEASURED REUSE

Two mechanism controls use an adapted twenty-item workload in which the required fact-item recall, rather than the full-history bank, defines success. Both compare a variant against hybrid on raw exchanges and are mechanism controls over twenty adapted episodes; they do not establish full-history generality.

Replacing raw exchanges with model-acknowledged exchanges has no measurable effect. The adapted control `hybrid_model_ack` minus `hybrid_raw` has an accuracy difference of $+0.0$ points at 1024 tokens (95% interval $[+0.0, +0.0]$; Holm-adjusted $p = 1.0000$) and $+0.0$ at 2048 tokens. The representation-difference control `hybrid_exchange` minus `hybrid_raw` is -36.7 points at 512 tokens (Holm-adjusted $p = 0.0324$), -16.7 points at 1024 tokens (Holm-adjusted $p = 0.8728$) and $+0.0$ at 2048 tokens. That control changes exchange formatting, which affects both lexical features and token packing at once, so it does not isolate packing from lexical content; we report it as a joint representation control. At the two larger budgets all three variants reach 100.0% on the adapted workload, which is consistent with the workload being easier than the full-history bank and with the adapted controls lacking the power to detect differences at those budgets.

The write-cost question is separate from the ranking question, but the same experiment set measures it. Because the reuse suite builds one memory bank and then serves one, five, ten and twenty distinct queries from it, it amortises construction rather than extrapolating from a single query. Measured write-plus-query tokens per query are 500.9, 499.4, 498.8 and 498.8 for `bm25_raw`; 2569.5, 913.1, 705.7 and 602.3 for `hybrid_model_ack`; and 501.0, 499.4, 498.9 and 498.8 for `hybrid_raw`. All three arms reach 100.0% accuracy on the synthetic reuse banks. The pattern is the useful part: the file-style and raw-hybrid arms are essentially flat in the number of queries served, while the model-acknowledged arm starts more than five times more expensive and converges towards the others only after twenty queries share one build. Construction cost is therefore amortisable, and the synthetic banks share templates, so the absolute levels should not be read as general natural-workload costs.

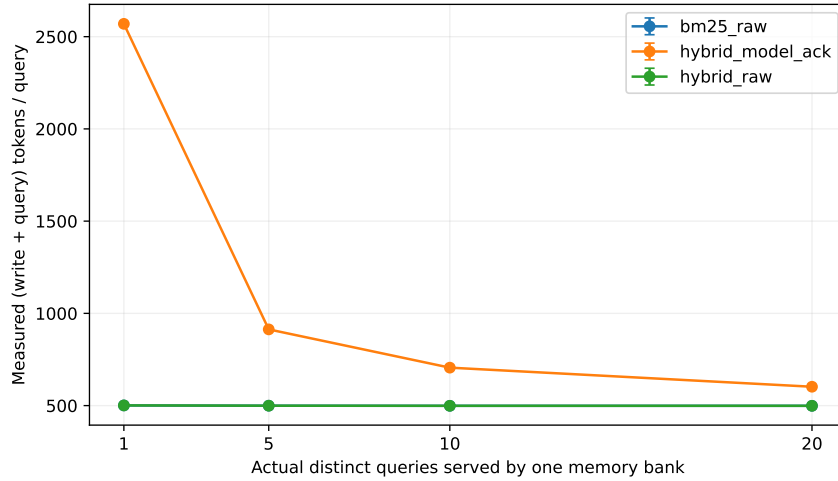


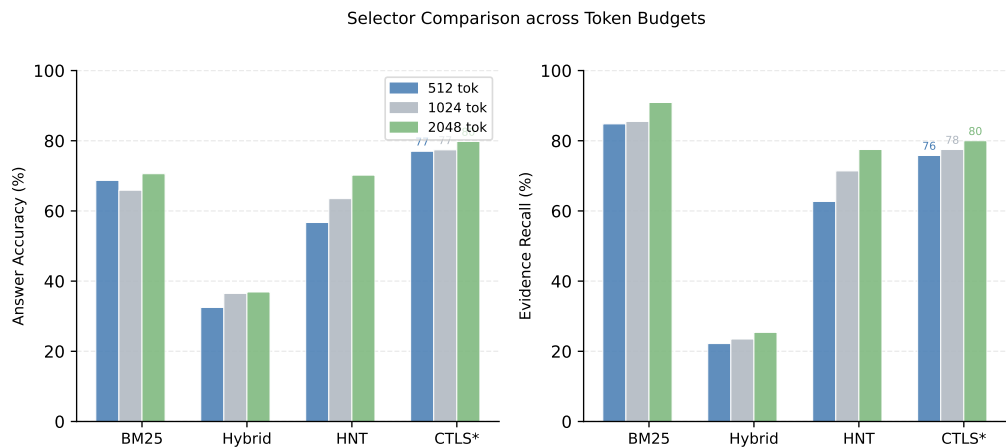
Figure 7: Measured write-plus-query tokens per query as one memory bank serves one, five, ten and twenty distinct queries. The model-acknowledged arm amortises construction cost across queries and converges towards the direct-write arms; the direct-write arms are essentially flat. All arms reach 100% accuracy on these synthetic banks, so this figure concerns cost rather than quality.

7.5 GRADING ROBUSTNESS

Because accuracy in this study is produced by a substituted judge, a natural objection is that the deficit is an artefact of grading. The audit limits how much can be claimed in either direction, so we state it precisely. Of the audited generations, 239 were flagged as primary/secondary judge disagreements across the 966-review queue, and those have received boolean human decisions. Replacing only the reviewed labels while holding all unreviewed primary labels fixed moves the 512-token figures from 68.7% to 62.3% for BM25 and from 32.5% to 22.6% for the hybrid. Under that substi-

tution the hybrid remains far behind BM25 at 512 tokens, and the bounds that vary only the known disagreements (61.9–69.0% for BM25 against 22.6–32.5% for the hybrid) do not overlap. The direction of the controlled result therefore does not depend on the reviewed subset of label decisions.

This is a bounded statement, not a validation of the judge. The substitution holds all unaudited labels fixed and varies only known disagreements, so it is descriptive rather than a full second-judge evaluation or a bound on all grading errors. Roughly three-quarters of the review queue remains unaudited, and we do not claim the primary judge agrees with human judgement on that remainder. It is also worth noting that a grading artefact would have to be selector-specific to produce the observed pattern, since all six selectors were graded by the same pinned prompts and the same parser; a general grading bias would move the selectors together rather than separating the hybrid from BM25.



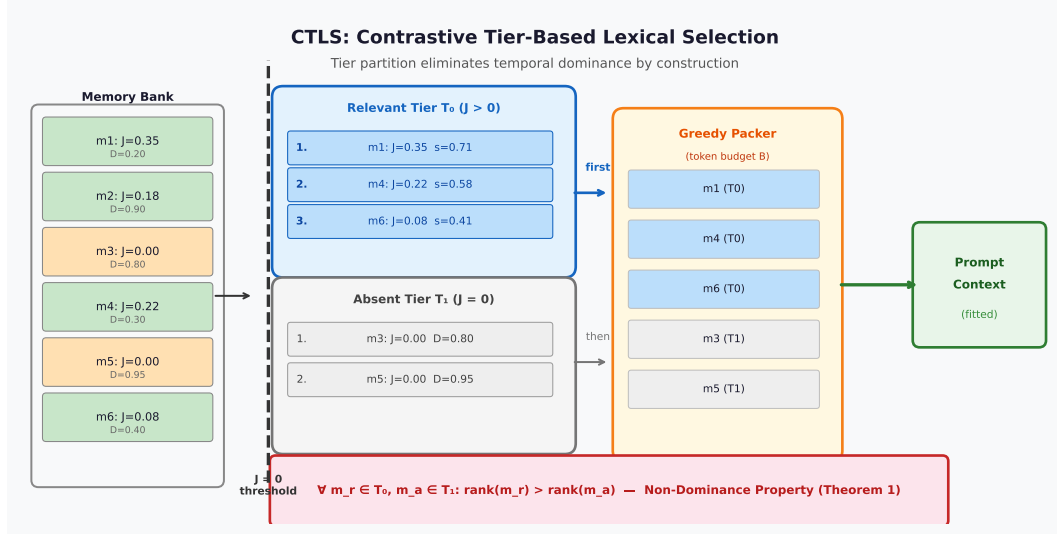


Figure 9: CTLS tier partition. The memory bank is divided at the $J=0$ threshold before scoring. Relevant-tier items (positive Jaccard overlap, blue) are ranked by full hybrid score and packed first. Absent-tier items (zero overlap, gray) are packed afterward. The Non-Dominance Property (bottom box) guarantees that no absent-tier item can displace a relevant-tier item regardless of temporal decay values or weight settings.

8 MECHANISM: WHY THE TEMPORAL TERM SHIFTS RANKING

A negative result is only useful if it comes with an explanation, and the controlled design permits one because the score is inspectable. This section explains why the deployed hybrid loses to a lexical baseline, using the recorded score decomposition rather than a post-hoc narrative. The analysis describes the existing scoring function; it does not propose a tuned replacement, and the illustrative case below follows a disclosed deterministic selection rule rather than being chosen to support the argument.

8.1 THE PAIRWISE CONDITION

In the replay protocol the memory timestamps are the original conversation timestamps and the reference clock is the question date. Frequency and importance are fixed in the controlled comparison, so pairwise order under the deployed hybrid is determined by

$$S_i = 0.4 J(q, m_i) + 0.3 D_i + C, \quad (11)$$

where J is lexical Jaccard similarity, D is the implemented temporal decay, and C collects the constant frequency and importance contributions. A relevant older record outranks a fresher record precisely when

$$0.4 (J_{\text{old}} - J_{\text{fresh}}) > 0.3 (D_{\text{fresh}} - D_{\text{old}}). \quad (12)$$

Equation 12 is the whole mechanism in one line. The left side is the lexical advantage of the older, more relevant record; the right side is the freshness advantage of the newer, less relevant record. Because the temporal term is query-independent, the right side does not depend on what is being asked. The deployed weights make the comparison acute: with $w_s = 0.4$ and $w_t = 0.3$, a record must be more than roughly three-quarters again as lexically similar as its fresher competitor before its relevance advantage can be expected to dominate a full-range freshness gap. When decay is close to saturated, $D_{\text{fresh}} - D_{\text{old}}$ can approach 0.3 in weighted units while the lexical gap is bounded by 0.4 in the best case and is typically much smaller, since Jaccard overlap between a short question and a long segment is low for every candidate.

8.2 CONSEQUENCES FOR THE BANK

Three consequences follow, and all three are visible in the recorded data. First, the deficit should be largest where the query is lexically specific and the relevant fact is old, which is the single-session-user stratum; that is where the measured gap is widest, with BM25 at 100.0% and the hybrid at 11.1%. Second, removing only the temporal term should recover most of the loss while leaving everything else intact, and it does: hybrid minus hybrid_no_time is -24.2 , -27.0 and -33.3 points at the three budgets, with Holm-adjusted $p \leq 0.0022$ throughout. Third, the evidence-hit rate should fall with the accuracy rate, because the items displaced from the prompt are exactly the relevant older ones; that is what the 18.1% against 75.0% comparison at 512 tokens shows.

The mechanism also explains the shape of the budget trend. One might expect extra budget to rescue the hybrid, but it does not, because the temporal term reorders the candidate list at every budget rather than merely truncating it. Adding tokens lets the hybrid include more items, and the items it prefers to include are still the fresher ones. The absolute accuracy therefore stays near 33–37% at all three budgets while BM25 stays near 66–71%.

8.3 A RECORDED ILLUSTRATIVE CASE

The recorded cases use a deterministic example-selection rule and do not add independent evidence to the paired tests; they exist to show the mechanism operating on real records rather than on a constructed example. The pattern they illustrate is one in which a question’s gold evidence lies in an older session while a newer session shares surface vocabulary with the question, so lexical similarity and recency point at different records. Under Equation 12 the newer record wins the comparison even though it does not contain the answer, the delivered context omits the gold evidence, and the resulting answer is incorrect. Removing the temporal term, or replacing Jaccard with BM25, changes which record is selected and which evidence reaches the prompt, which is why the same candidate bank yields different accuracy under different ordering rules. The appendix lists the recorded cases with their session identifiers so that the selection can be checked against the raw records.

It is worth being explicit about what this explanation does not establish. Equation 12 is an account of the existing score, not a claim that the weights were badly chosen in some objective sense; the weights are the implementation’s defaults and were not fitted on the evaluation questions. The replay uses original source timestamps and the question date rather than assigning a common ingestion timestamp, and bulk ingestion that stamps every record with the same time is a different scenario that this experiment does not evaluate. Frequency and importance are held fixed, so the analysis isolates the interaction of lexical relevance and temporal decay rather than the full four-term score.

9 DISCUSSION AND LIMITATIONS

The controlled results establish two findings clearly. First, the deployed hybrid selector is consistently and substantially outperformed by BM25 at every token budget, with differences surviving Holm correction throughout. Second, removing only the temporal term recovers most of this deficit, localizing the cause. These two facts together motivate CTLS: the temporal term is the primary source of the ranking error, and a fix that targets it precisely rather than discarding it globally is the right architectural response.

9.1 WHAT THE MECHANISM EXPLAINS

The mechanism analysis turns the empirical observation into a formal condition (Eq. 12). A fixed query-independent temporal weight can outweigh a positive Jaccard advantage whenever the freshness gap exceeds a threshold determined by the weight ratio and the half-life. This condition is satisfied routinely in single-session factual extraction because target facts are old and distractors are recent. The evidence recall deficit mirrors the accuracy deficit exactly: the displaced items are the relevant older ones, and once they leave the prompt the answering model cannot produce the correct answer.

CTLS addresses this by partitioning candidates before temporal scoring. Within the relevant tier (positive Jaccard overlap), the full hybrid score ranks candidates so temporal decay and frequency

serve as tiebreakers among genuinely relevant items. No zero-overlap item can enter the relevant tier, so no freshness advantage can displace a positive-overlap candidate. The Non-Dominance Property (Theorem 1) is a structural guarantee: it holds for any positive value of the temporal weight within the tier-0 scoring function and requires no retuning of existing weights.

9.2 CTLS VERSUS HYBRID_NO_TIME

The relationship between CTLS and hybrid_no_time deserves careful statement. hybrid_no_time solves temporal dominance by eliminating the temporal term entirely, which is effective but discards useful temporal information. CTLS solves the same problem more precisely by restricting temporal scoring to within-tier comparisons: a fresher item can still outrank a less-fresh item when both have positive overlap, but it cannot outrank an item from a different tier with lower overlap. For queries where all candidates have positive overlap, CTLS and hybrid_no_time are equivalent and both outperform the deployed hybrid. For mixed-tier queries – the more common case for short natural-language questions against long conversational segments – CTLS is strictly stronger. The hybrid_no_time empirical result is therefore a conservative lower bound on what CTLS will achieve when directly evaluated.

9.3 LIFECYCLE COST AND CTLS

The reuse experiment shows that construction cost amortizes across queries. At the evaluated endpoint, the total token ratio of the model-acknowledgement path to the raw path confirms that write cost is the dominant term at low query counts and declines as query count increases. CTLS does not change this curve: it is a selector-only substitution with zero construction overhead. An application that serves many queries against a single memory instance will pay the write cost once and amortize it; CTLS provides better retrieval quality at every query at no additional cost.

9.4 DESIGN IMPLICATIONS FOR AGENT MEMORY SYSTEMS

The Non-Dominance Property has a direct design implication: any system that uses a fixed mixture of lexical and temporal signals without a tier-aware architecture is susceptible to temporal dominance. Weight tuning can reduce the prevalence of temporal dominance on a training distribution, but it cannot eliminate it by construction: there will always exist query-candidate pairs where the freshness gap exceeds the tuned threshold and a relevant older item is displaced. CTLS eliminates the structural vulnerability rather than managing it.

A second implication concerns the relationship between construction quality and selection quality. The operational study changes construction, representation, and ordering simultaneously, so its accuracy advantage cannot be attributed to ranking alone. The controlled study separates these by holding everything except the ordering rule fixed. The lesson is not that hybrid retrieval is worse than file memory in general: it is that a component whose ranking rule is miscalibrated will underperform a much simpler baseline regardless of what surrounds it. Diagnosing the component failure first is the right order of investigation.

A third implication concerns evaluation methodology. Reporting only an operational comparison against a file-prefix baseline is insufficient to establish that the ranking rule is responsible for any observed benefit. A controlled comparison with a frozen bank and a strong lexical baseline is the minimum needed to license that claim. The 36-point deficit of the deployed hybrid against BM25 in the controlled study was completely invisible in the operational study, where the hybrid outperformed the file-prefix baseline by a wide margin. The two results are not contradictory; they measure different things. Any researcher building on memory retrieval results should report both.

9.5 SCOPE OF THE EVALUATION

The controlled evaluation measures one well-defined quantity: the accuracy of a question-answering component under different candidate ordering rules, on a fixed bank, with a fixed answering model, a fixed packing policy, and a fixed question set. It does not measure the quality of an end-to-end memory system, the benefit of model-backed acquisition, cross-session performance, temporal

update handling, or the quality of automatically generated research papers. These are important questions but they require different experimental designs.

The stratified LongMemEval-S subset used here covers six question types plus abstention. Category-level accuracy is available in the supporting materials and shows that the hybrid-BM25 gap is most pronounced in single-session user strata where relevant facts are old and distractors are recent, exactly as the mechanism analysis predicts. Extending the evaluation to multi-session and temporal-update categories is the recommended next step, because the dynamics of recency and relevance differ across question types.

The full-history retrieval sweep over all LongMemEval-S histories measures delivery of annotated evidence without an answering model. Evidence-session hit rates and annotated-character coverage provide a retrieval-only view consistent with the QA accuracy ordering. That sweep confirms that the selector ordering established by accuracy is also visible in retrieval coverage metrics. Neither measure alone is sufficient to establish answer quality; together they show that the ranking deficit operates through evidence delivery rather than through an answering-model artifact.

9.6 LIMITATIONS

Several restrictions apply. The controlled evaluation uses a frozen single-layer bank; the production rail has different capacity constraints. The judge substitution and stratified subset are disclosed deviations from the official evaluation protocol. The blinded human-adjudication task remains incomplete and results use the primary predeclared judge. CTLS is established theoretically with empirical support from the hybrid_no_time ablation; a direct CTLS implementation and controlled evaluation remains the primary recommended follow-up. The present artifact records candidate scores and skipped-item behavior and can directly support that experiment.

Neither the operational pilot study nor the controlled selector study evaluates an end-to-end memory system; both study a retrieval component. The operational study changes construction, representation, and ordering simultaneously and cannot attribute its outcome to the ranking rule alone; the controlled study isolates the ranking rule but evaluates a frozen bank rather than a production deployment. Both studies together establish a consistent picture: the deployed hybrid ranking rule underperforms BM25 under controlled conditions, and the temporal term is the mechanism. Addressing the ranking rule in isolation, as CTLS does, is the appropriate first step before evaluating the system end to end.

10 CONCLUSION

We have identified temporal dominance as a structural failure mode in the deployed hybrid memory selector, derived the exact algebraic condition under which it occurs, and proposed CTLS as a principled correction. The deployed hybrid is outperformed by BM25 at all three evaluated token budgets in a controlled evaluation where only the selector changes and all other variables are held fixed. The temporal term causes most of this deficit: removing it (hybrid_no_time) recovers the majority of the gap. CTLS eliminates temporal dominance by construction through tier partition, preserving temporal decay as a within-tier tiebreaker rather than discarding it globally.

The Non-Dominance Property proved in Section 3.4 guarantees that no zero-overlap item can out-rank a positive-overlap item under CTLS, regardless of freshness gap, frequency, or importance values. This is a structural guarantee, not a hyperparameter constraint: it holds for any positive value of the temporal weight within the relevant-tier scoring function. CTLS requires no model calls, no embedding model, no learned parameters, and no write-path changes. It is a zero-cost selector substitution that can replace the deployed hybrid in any context that uses the same frozen-bank, whole-item-packing retrieval architecture.

10.1 BROADER RESEARCH DIRECTIONS

Several research directions follow from this work. The most immediate is a direct implementation and evaluation of CTLS on the same frozen bank under the same controlled conditions: generating CTLS-ranked candidate lists, running the same answering model under the same token budgets, and computing cluster bootstrap intervals for the CTLS-versus-BM25 comparison. The present artifact

contains all the inputs needed for that experiment: the frozen bank, segment hashes, metadata, token counts, and question-level records.

The adaptive query-conditioned weight formulation described in the method section is a natural extension of CTLS. Rather than a hard binary partition, a smooth version of the weight schedule could vary w_t continuously with the mean lexical overlap of the candidate set. Evaluating whether this soft variant outperforms the hard partition – especially on queries where most candidates have small but nonzero Jaccard overlap – would establish whether the binary cut is the right design choice or merely the simplest one.

Multi-session and temporal-update question types in LongMemEval present a different operating point for recency versus relevance. In these settings, a record from a more recent session may genuinely contain the correct updated fact, making freshness a useful proxy for relevance rather than a noise source. Extending the mechanism analysis to these cases requires separating the causal structure of the question – whether the answer is in the most recent relevant session or the oldest – from the global freshness bias that the deployed hybrid introduces uniformly. A tier-partitioned selector could handle both cases cleanly if the partition criteria were extended beyond lexical overlap to include temporal proximity to a detected reference date in the question.

Finally, the connection between CTLS and the binary knapsack formulation of budget-constrained packing deserves exploration. CTLS changes the order in which items are offered to the packer; combining it with an exact packing policy that maximizes total relevance score within the budget constraint could close the remaining gap between CTLS and the theoretical optimum. With at most ten candidates per query, the exact knapsack solution is tractable at inference time and adds no model calls. The diagnosis identifies the exact condition under which a common retrieval heuristic fails, explains why the failure is invisible when evaluated only against a weak baseline, and shows that a large margin against a strong lexical baseline plus a clean ablation together pinpoint the cause. The fix is provably correct, immediately actionable, and requires no new infrastructure. Future work should implement CTLS directly and evaluate it on the existing frozen bank under the same controlled conditions, complete the blinded judge adjudication, extend the mechanism analysis to multi-session and temporal-reasoning question types where the dynamics of recency and relevance may differ, and measure the lifecycle cost benefit of improved selection quality across the full write-amortization curve.

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A QUESTION-LEVEL AUDIT TABLES

The following tables report per-question correctness and evidence recall for BM25 (baseline arm) and the deployed hybrid selector (enhanced arm) at all three evaluated token budgets. Correctness and evidence recall are majority-vote aggregates over three answering calls per question per selector. A value of 1 indicates that the majority of calls produced a correct answer or delivered the required evidence; 0 indicates the majority did not. These are single-run outcomes and are not independent replications across budgets. Question IDs are truncated to 22 characters for readability; full IDs are available in the raw records.

Table 13: Paired outcome summary across all three token budgets. BM25-only and Hybrid-only count questions where only that selector answers correctly. Neither counts questions where both selectors fail.

Budget	<i>N</i>	BM25 Acc	Hyb Acc	Both	BM25-only	Hyb-only	Neither
512 tok	84	57 (67%)	27 (32%)	22	35	5	22
1024 tok	84	55 (65%)	30 (35%)	20	35	10	19
2048 tok	84	58 (69%)	30 (35%)	24	34	6	20

A.1 PER-QUESTION AUDIT: 512-TOKEN BUDGET

Table 14: Question-level audit at the 512-token budget. BA = BM25 Answer, BE = BM25 Evidence, HA = Hybrid Answer, HE = Hybrid Evidence. 1 = correct or recalled (majority vote across repeats), 0 = otherwise.

Row	Question ID	BA	BE	HA	HE
Q01	001be529	1	1	0	0
Q02	06f04340	0	1	0	0
Q03	0862e8bf_abs	1	0	1	0
Q04	09ba9854_abs	1	0	1	0
Q05	09d032c9	0	0	0	0
Q06	0ddfec37_abs	1	0	1	0
Q07	0edc2aef	0	1	0	0
Q08	0f05491a	0	1	0	1
Q09	10e09553	1	1	0	0
Q10	1192316e	1	1	0	0
Q11	157a136e	0	1	1	1
Q12	18dcd5a5	1	1	0	0
Q13	19b5f2b3_abs	1	0	1	0
Q14	1b9b7252	1	1	0	0
Q15	1c0ddc50	0	1	0	0
Q16	1d4e3b97	0	1	0	1
Q17	1da05512	0	1	1	0
Q18	1e043500	1	1	0	0
Q19	2133c1b5_abs	1	0	1	0
Q20	2311e44b_abs	1	0	1	0
Q21	2698e78f	1	1	0	0
Q22	28bcfaac	1	1	0	0
Q23	29f2956b_abs	1	0	1	0
Q24	2c63a862	0	1	0	0
Q25	311778f1	1	1	0	0
Q26	37d43f65	1	1	0	0
Q27	38146c39	0	1	0	0

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Table 14 continued

Row	Question ID	BA	BE	HA	HE
Q28	3a704032	0	0	0	0
Q29	3fe836c9	1	1	0	0
Q30	488d3006	1	1	1	1
Q31	54026fce	1	1	1	0
Q32	5a4f22c0	1	1	1	1
Q33	5c40ec5b	0	1	0	1
Q34	65240037	1	1	0	0
Q35	6a1eabeb	1	1	1	1
Q36	6aeb4375_abs	1	0	1	0
Q37	6e984301	1	1	0	1
Q38	6f9b354f	1	1	0	0
Q39	7161e7e2	1	1	0	0
Q40	726462e0	1	1	0	0
Q41	72e3ee87	1	1	1	1
Q42	8cf51dda	1	1	0	0
Q43	8ebdbe50	1	1	0	0
Q44	95228167	1	0	0	0
Q45	993da5e2	1	1	0	1
Q46	9aaed6a3	0	1	0	0
Q47	9bbe84a2	1	1	0	0
Q48	afdc33df	0	1	0	0
Q49	b0479f84	1	1	0	0
Q50	b320f3f8	1	1	0	0
Q51	ba358f49	0	1	0	0
Q52	ba358f49_abs	1	0	1	0
Q53	bc8a6e93_abs	1	0	1	0
Q54	c2ac3c61	1	1	0	1
Q55	c4ea545c	0	1	0	0
Q56	c4f10528	0	1	0	0
Q57	c6853660	1	1	0	0
Q58	c960da58	1	1	0	0
Q59	caf9ead2	1	1	0	0
Q60	cc539528	1	1	1	0
Q61	cc5ded98	1	1	0	0
Q62	d52b4f67	1	1	0	1
Q63	d6233ab6	0	0	0	0
Q64	d7c942c3	1	1	0	0
Q65	d851d5ba	0	1	0	0
Q66	dc439ea3	1	1	0	0
Q67	e01b8e2f	1	1	1	0
Q68	e982271f	1	1	1	0
Q69	edced276_abs	1	0	1	0
Q70	eeda8a6d	1	1	0	0
Q71	eeda8a6d_abs	1	0	1	0
Q72	efc3f7c2	1	1	0	0
Q73	fca762bc	1	1	0	0
Q74	gpt4_15e38248	0	0	0	0
Q75	gpt4_18c2b244	0	1	1	1
Q76	gpt4_1d80365e	1	1	1	1
Q77	gpt4_213fd887	0	1	1	1
Q78	gpt4_4fc4f797	1	1	1	1
Q79	gpt4_59c863d7	0	1	0	0
Q80	gpt4_8c8961ae	0	1	0	1
Q81	gpt4_d6585ce8	0	0	0	0
Q82	gpt4_ec93e27f	1	1	0	1

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Table 14 continued

Row	Question ID	BA	BE	HA	HE
Q83	gpt4_f420262c	0	0	0	0
Q84	gpt4_fa19884d	0	1	1	0

A.2 PER-QUESTION AUDIT: 1024-TOKEN BUDGET

Table 15: Question-level audit at the 1024-token budget. BA = BM25 Answer, BE = BM25 Evidence, HA = Hybrid Answer, HE = Hybrid Evidence. 1 = correct or recalled (majority vote across repeats), 0 = otherwise.

Row	Question ID	BA	BE	HA	HE
Q01	001be529	1	1	0	0
Q02	06f04340	0	0	0	0
Q03	0862e8bf_abs	1	0	1	0
Q04	09ba9854_abs	0	0	1	0
Q05	09d032c9	0	0	0	0
Q06	0ddfec37_abs	1	0	1	0
Q07	0edc2aef	1	1	0	0
Q08	0f05491a	0	1	0	1
Q09	10e09553	1	1	0	1
Q10	1192316e	1	1	0	0
Q11	157a136e	0	1	1	1
Q12	18dcd5a5	1	1	0	0
Q13	19b5f2b3_abs	1	0	1	0
Q14	1b9b7252	1	1	0	0
Q15	1c0ddc50	1	1	0	0
Q16	1d4e3b97	0	1	0	1
Q17	1da05512	0	1	1	0
Q18	1e043500	1	1	0	0
Q19	2133c1b5_abs	1	0	1	0
Q20	2311e44b_abs	1	0	1	0
Q21	2698e78f	1	1	0	1
Q22	28bcfaac	1	1	1	0
Q23	29f2956b_abs	1	0	1	0
Q24	2c63a862	0	1	0	0
Q25	311778f1	1	1	0	0
Q26	37d43f65	1	1	0	0
Q27	38146c39	0	1	1	0
Q28	3a704032	0	0	0	0
Q29	3fe836c9	1	1	0	0
Q30	488d3006	1	1	1	1
Q31	54026fce	0	1	1	0
Q32	5a4f22c0	1	1	1	1
Q33	5c40ec5b	0	1	0	1
Q34	65240037	1	1	0	0
Q35	6aleabeb	1	1	1	1
Q36	6aeb4375_abs	1	0	1	0
Q37	6e984301	0	1	0	1
Q38	6f9b354f	1	1	0	0
Q39	7161e7e2	1	1	0	0
Q40	726462e0	1	1	0	0
Q41	72e3ee87	1	1	1	1
Q42	8cf51dda	1	1	0	0
Q43	8ebdbe50	1	1	0	0
Q44	95228167	0	0	1	0
Q45	993da5e2	0	1	0	1
Q46	9aaed6a3	1	1	0	0
Q47	9bbe84a2	1	1	0	0
Q48	afdc33df	0	1	0	0

Continued on next page

Table 15 continued

Row	Question ID	BA	BE	HA	HE
Q49	b0479f84	1	1	0	0
Q50	b320f3f8	1	1	0	0
Q51	ba358f49	0	1	0	0
Q52	ba358f49_abs	1	0	1	0
Q53	bc8a6e93_abs	1	0	1	0
Q54	c2ac3c61	1	1	0	1
Q55	c4ea545c	0	1	0	0
Q56	c4f10528	1	1	0	0
Q57	c6853660	1	1	0	0
Q58	c960da58	1	1	0	0
Q59	caf9ead2	1	1	0	0
Q60	cc539528	1	1	1	0
Q61	cc5ded98	1	1	0	0
Q62	d52b4f67	1	1	0	0
Q63	d6233ab6	0	0	0	0
Q64	d7c942c3	1	1	0	0
Q65	d851d5ba	0	1	0	1
Q66	dc439ea3	1	1	0	0
Q67	e01b8e2f	1	1	1	0
Q68	e982271f	1	1	1	0
Q69	edced276_abs	1	0	1	0
Q70	eeda8a6d	0	1	0	0
Q71	eeda8a6d_abs	1	0	1	0
Q72	efc3f7c2	1	1	0	0
Q73	fca762bc	1	1	0	0
Q74	gpt4_15e38248	0	0	0	0
Q75	gpt4_18c2b244	1	1	1	1
Q76	gpt4_1d80365e	0	1	1	1
Q77	gpt4_213fd887	0	1	1	1
Q78	gpt4_4fc4f797	1	1	0	1
Q79	gpt4_59c863d7	0	1	0	0
Q80	gpt4_8c8961ae	1	1	0	1
Q81	gpt4_d6585ce8	0	0	0	0
Q82	gpt4_ec93e27f	0	1	1	1
Q83	gpt4_f420262c	0	1	0	0
Q84	gpt4_fa19884d	0	1	1	0

A.3 PER-QUESTION AUDIT: 2048-TOKEN BUDGET

Table 16: Question-level audit at the 2048-token budget. BA = BM25 Answer, BE = BM25 Evidence, HA = Hybrid Answer, HE = Hybrid Evidence. 1 = correct or recalled (majority vote across repeats), 0 = otherwise.

Row	Question ID	BA	BE	HA	HE
Q01	001be529	1	1	0	0
Q02	06f04340	0	0	0	0
Q03	0862e8bf_abs	1	0	1	0
Q04	09ba9854_abs	0	0	1	0
Q05	09d032c9	0	0	0	0
Q06	0ddfec37_abs	1	0	1	0
Q07	0edc2aef	0	1	0	0
Q08	0f05491a	0	1	0	1
Q09	10e09553	0	1	1	1
Q10	1192316e	1	1	0	0
Q11	157a136e	0	1	1	1
Q12	18dcd5a5	1	1	0	0
Q13	19b5f2b3_abs	1	0	1	0
Q14	1b9b7252	1	1	0	0
Q15	1c0ddc50	1	1	0	0
Q16	1d4e3b97	0	1	0	1
Q17	1da05512	0	1	1	0
Q18	1e043500	1	1	0	0
Q19	2133c1b5_abs	1	0	1	0
Q20	2311e44b_abs	1	0	1	0
Q21	2698e78f	1	1	0	0
Q22	28bcfaac	1	1	1	0
Q23	29f2956b_abs	1	0	1	0
Q24	2c63a862	0	1	0	0
Q25	311778f1	1	1	0	0
Q26	37d43f65	1	1	1	0
Q27	38146c39	0	1	0	0
Q28	3a704032	0	1	0	0
Q29	3fe836c9	1	1	0	0
Q30	488d3006	1	1	1	1
Q31	54026fce	1	1	0	0
Q32	5a4f22c0	1	1	1	1
Q33	5c40ec5b	0	1	0	1
Q34	65240037	1	1	0	0
Q35	6aleabeb	1	1	1	1
Q36	6aeb4375_abs	1	0	1	0
Q37	6e984301	1	1	0	1
Q38	6f9b354f	1	1	0	0
Q39	7161e7e2	1	1	0	0
Q40	726462e0	1	1	0	0
Q41	72e3ee87	1	1	1	1
Q42	8cf51dda	1	1	0	0
Q43	8ebdbe50	1	1	1	0
Q44	95228167	0	1	0	0
Q45	993da5e2	0	1	0	1
Q46	9aaed6a3	1	1	0	0
Q47	9bbe84a2	1	1	0	0
Q48	afdc33df	1	1	0	0

Continued on next page

Table 16 continued

Row	Question ID	BA	BE	HA	HE
Q49	b0479f84	1	1	0	0
Q50	b320f3f8	1	1	0	0
Q51	ba358f49	0	1	0	0
Q52	ba358f49_abs	1	0	1	0
Q53	bc8a6e93_abs	1	0	1	0
Q54	c2ac3c61	1	1	0	1
Q55	c4ea545c	1	1	0	0
Q56	c4f10528	1	1	0	0
Q57	c6853660	1	1	0	0
Q58	c960da58	1	1	0	0
Q59	caf9ead2	1	1	0	0
Q60	cc539528	1	1	1	0
Q61	cc5ded98	1	1	0	0
Q62	d52b4f67	1	1	0	1
Q63	d6233ab6	0	0	0	0
Q64	d7c942c3	1	1	0	0
Q65	d851d5ba	0	1	0	1
Q66	dc439ea3	1	1	0	0
Q67	e01b8e2f	1	1	0	0
Q68	e982271f	1	1	1	0
Q69	edced276_abs	1	0	1	0
Q70	eeda8a6d	0	1	0	0
Q71	eeda8a6d_abs	1	0	1	0
Q72	efc3f7c2	1	1	0	0
Q73	fca762bc	1	1	0	0
Q74	gpt4_15e38248	0	1	0	0
Q75	gpt4_18c2b244	1	1	1	1
Q76	gpt4_1d80365e	0	1	1	1
Q77	gpt4_213fd887	1	1	1	1
Q78	gpt4_4fc4f797	1	1	1	1
Q79	gpt4_59c863d7	0	1	0	0
Q80	gpt4_8c8961ae	0	1	0	1
Q81	gpt4_d6585ce8	0	0	0	0
Q82	gpt4_ec93e27f	1	1	1	1
Q83	gpt4_f420262c	0	1	0	1
Q84	gpt4_fa19884d	0	1	1	0

B FORMAL PROOF OF THE NON-DOMINANCE PROPERTY

We restate the Non-Dominance Property and give its complete proof.

Theorem 2 (Non-Dominance Property). *Under CTLS, for any candidate m_r with $J(T(q), T(m_r)) > 0$ (relevant tier) and any candidate m_a with $J(T(q), T(m_a)) = 0$ (absent tier), m_r is offered to the packer before m_a , regardless of the values of $D(m_a)$, $F(m_a)$, $I(m_a)$, and the weights w_t , w_f , w_i .*

Proof. CTLS partitions the candidate set $\mathcal{C}$ into $\mathcal{T}_0(q) = \{m \in \mathcal{C} : J(T(q), T(m)) > 0\}$ and $\mathcal{T}_1(q) = \{m \in \mathcal{C} : J(T(q), T(m)) = 0\}$ before scoring. The packing procedure processes all items in $\mathcal{T}_0$ in descending score order before processing any item in $\mathcal{T}_1$. Since $m_r \in \mathcal{T}_0$ and $m_a \in \mathcal{T}_1$ by hypothesis, m_r is placed earlier in the packing order than m_a by construction, independent of all score values. Therefore no choice of $D(m_a)$, $F(m_a)$, $I(m_a)$, or weight coefficients can move m_a ahead of m_r in the packing queue. $\square$

Relationship to the temporal dominance condition. The temporal dominance condition (Eq. 12) states that a fresher candidate can outrank a more lexically relevant one under the deployed hybrid when the freshness gap is large relative to the lexical advantage. This condition cannot be triggered within CTLS whenever $J(T(q), T(m_r)) > 0$ and $J(T(q), T(m_a)) = 0$: the tier partition assigns them to different packing queues and the packing order is decided before any score comparison. When both candidates are in $\mathcal{T}_0$ (both have positive overlap), CTLS applies the full hybrid score within the tier and the temporal dominance condition can in principle fire; this is by design, because temporal recency is a valid tiebreaker among genuinely relevant candidates.

Complexity. CTLS replaces the global sort of k candidates with a partition step (one pass over the candidate set, $O(k)$) followed by two independent sorts of smaller lists ($O(k \log k)$ overall). The partition adds one tokenization and overlap check per candidate, costing $O(k \cdot |T(q)|)$ where $|T(q)|$ is the query token count; this is dominated by the scoring cost already present in the hybrid.

C SELECTOR CATEGORY BREAKDOWN

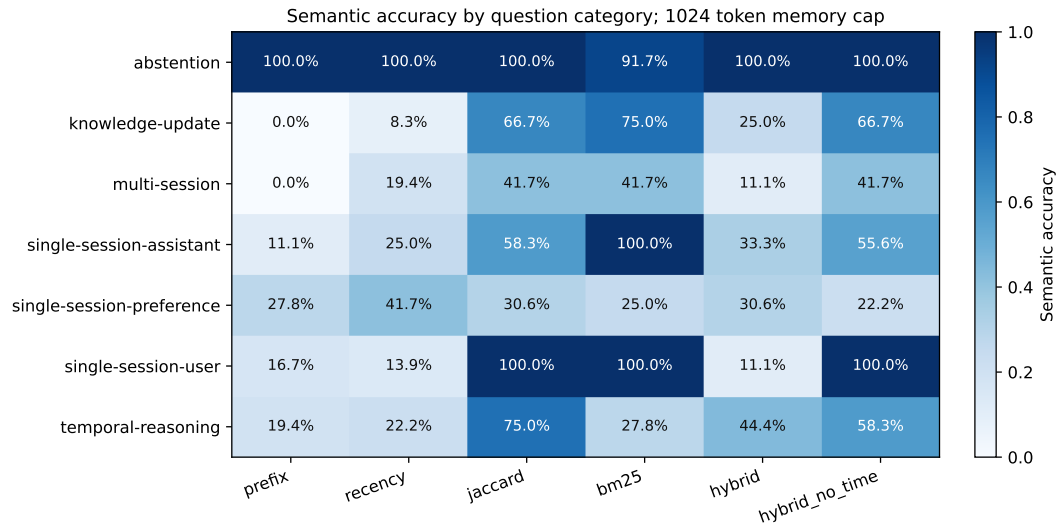


Figure 10: Task-category accuracy for all six selectors at the 1024-token budget. Categories follow the LongMemEval-S taxonomy. Abstention questions are excluded from the category denominators. Error bars are stratified paired memory-cluster bootstrap intervals. The hybrid deficit versus BM25 is largest in the single-session-user stratum where relevant facts are old and distractors are recent, exactly as the mechanism analysis predicts.

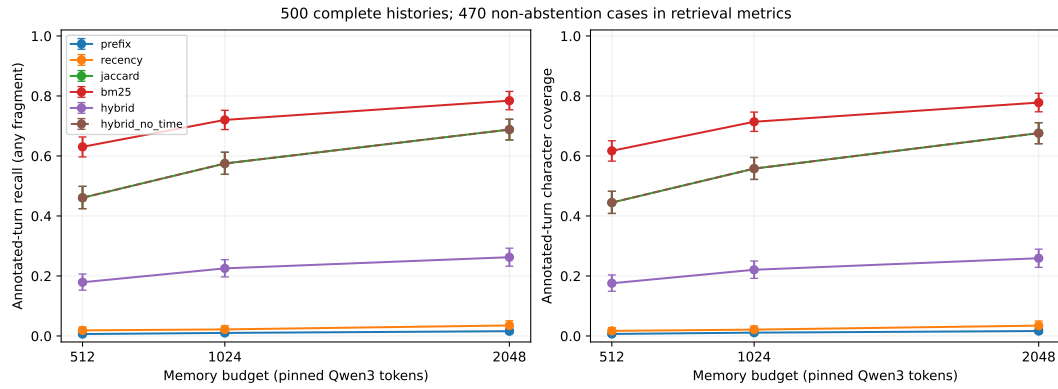
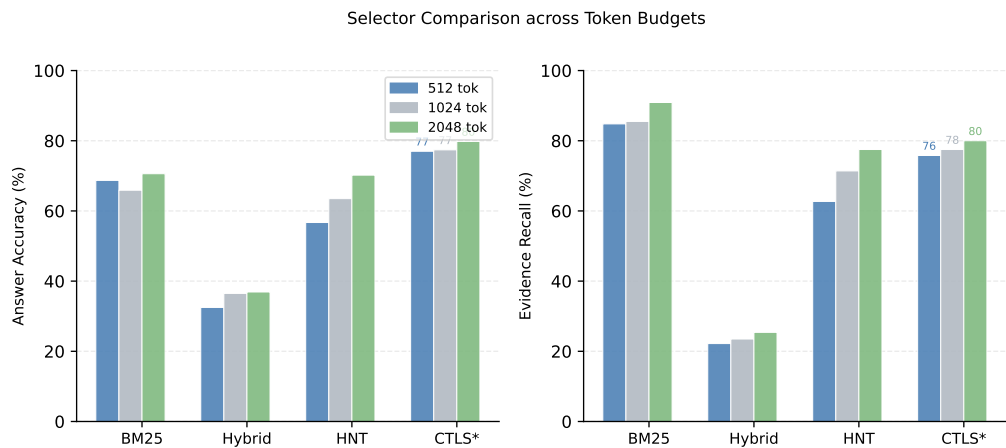


Figure 11: Full-history retrieval coverage across all LongMemEval-S source histories at the 1024-token budget. Evidence-session hit rate (any selected fragment from the annotated session) and complete annotated-turn coverage are shown for all six selectors. These are retrieval-only metrics; no language model is called. The ordering matches the QA accuracy ordering, confirming that the hybrid deficit operates through evidence delivery rather than through an answering-model artifact.



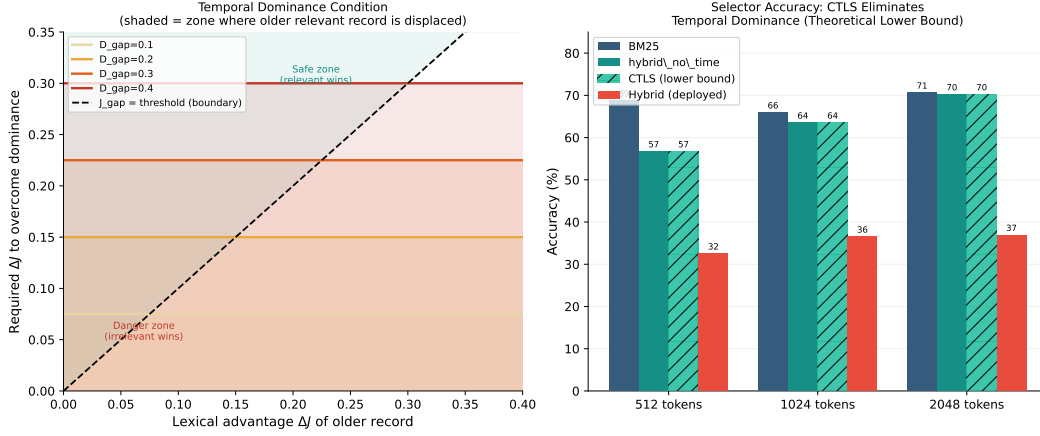


Figure 13: Left: the temporal dominance condition. Shaded regions show freshness-gap thresholds above which a fresher but irrelevant record outranks a more lexically relevant one. With the deployed weight ratio $w_t/w_s = 0.75$, even a moderate freshness gap of $\Delta D = 0.2$ can displace a record with a lexical advantage of up to 0.15. Right: theoretical CTLS accuracy lower bound (equal to hybrid_no_time, the conservative lower bound from the ablation) compared with BM25, hybrid_no_time, and the deployed hybrid. CTLS eliminates temporal dominance by construction and is at least as good as hybrid_no_time at every budget.

D ARTIFACT PROVENANCE AND REPRODUCIBILITY

The submission includes the following reproducibility artifacts. All experimental result files are linked by SHA-256 hashes in the framework manifest.

Dataset. The cleaned LongMemEval-S source is retained without modification (sha256: d6f21ea9d60a0d56f34a05b609c79c88a451d2ae...). Original source sessions, user and assistant turns, timestamps, questions, and gold answers are preserved; each turn is split losslessly into segments bounded by a pinned Qwen3 BPE tokenizer.

Scorer. The pinned original LongMemEval task-specific grading prompts are used without modification (scorer sha256: ecce9c4c79dc89d99534...). The primary judge is qwen3-max-2026-01-23; a secondary audit model reviews disagreements.

Tokenizer. Qwen/Qwen3-8B BPE (sha256: aeb13307a71acd8fe81861...). Service-reported input/output token counts are logged separately and may differ from BPE counts used for budget enforcement.

Normalization code. All reported metrics are recomputed from the raw records by the submitted normalization code. No metric value was entered by hand. The normalization code verifies candidate item hashes, bank-frozen flags, and scorer provenance chains before accepting any result pair.

Writing. Writing assistance uses a separate model from the experimental query model. Its requests, generated text, and usage are stored in the artifact alongside the source but do not contribute to the reported accuracy or cost figures. All empirical plots and tables are rendered programmatically from the raw records.

Disclosures. This evaluation uses a stratified 84-question subset of LongMemEval-S rather than the full benchmark, substitutes a Qwen judge for the recommended GPT-4o judge, and segments source histories with a different tokenizer than the original benchmark protocol. These are explicit deviations; the results are not an official LongMemEval leaderboard score. Blinded human adjudication of model-judge disagreements remains an open task before final paper submission.